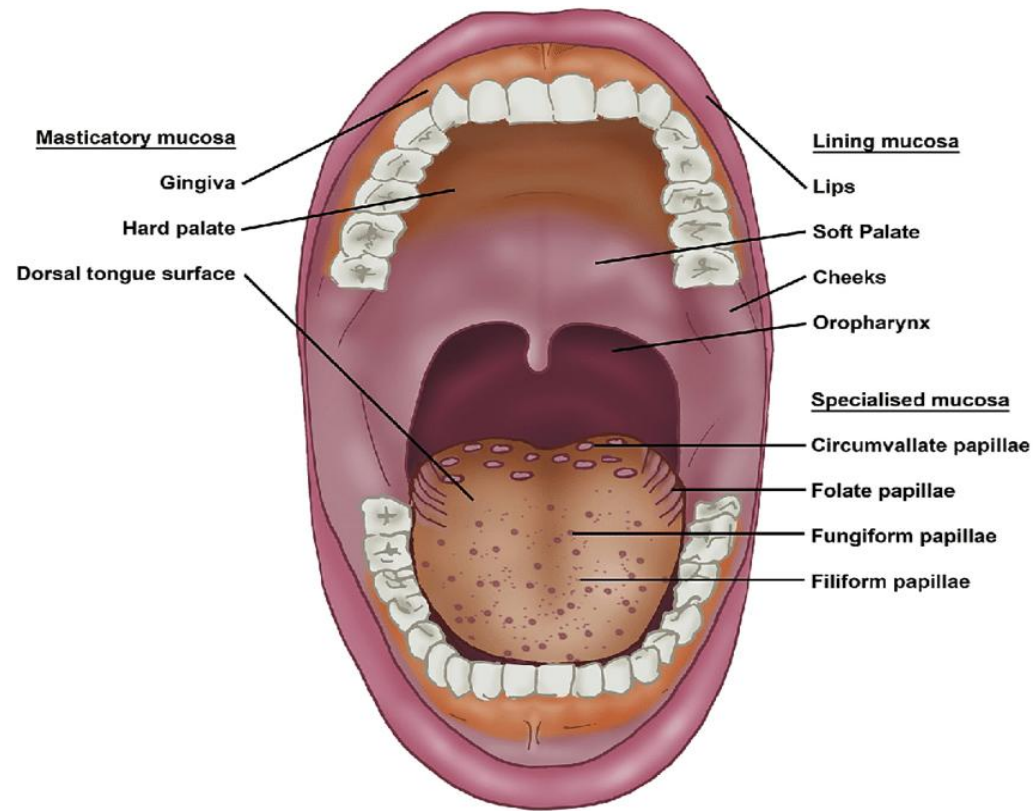


ORAL MUCOUS MEMBRANE

Part-2



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COURSE CO-ORDINATOR
ORAL CAVITY IN HEALTH

Lecture Learning Outcomes: L-12 OMM-2,3

By the end of this lesson students will be able to:

12.1: Illustrate the layers of stratum squamous epithelium

12.2: Describe the non keratinized and keratinized epithelium

12.3: Discuss the non keratinocytes

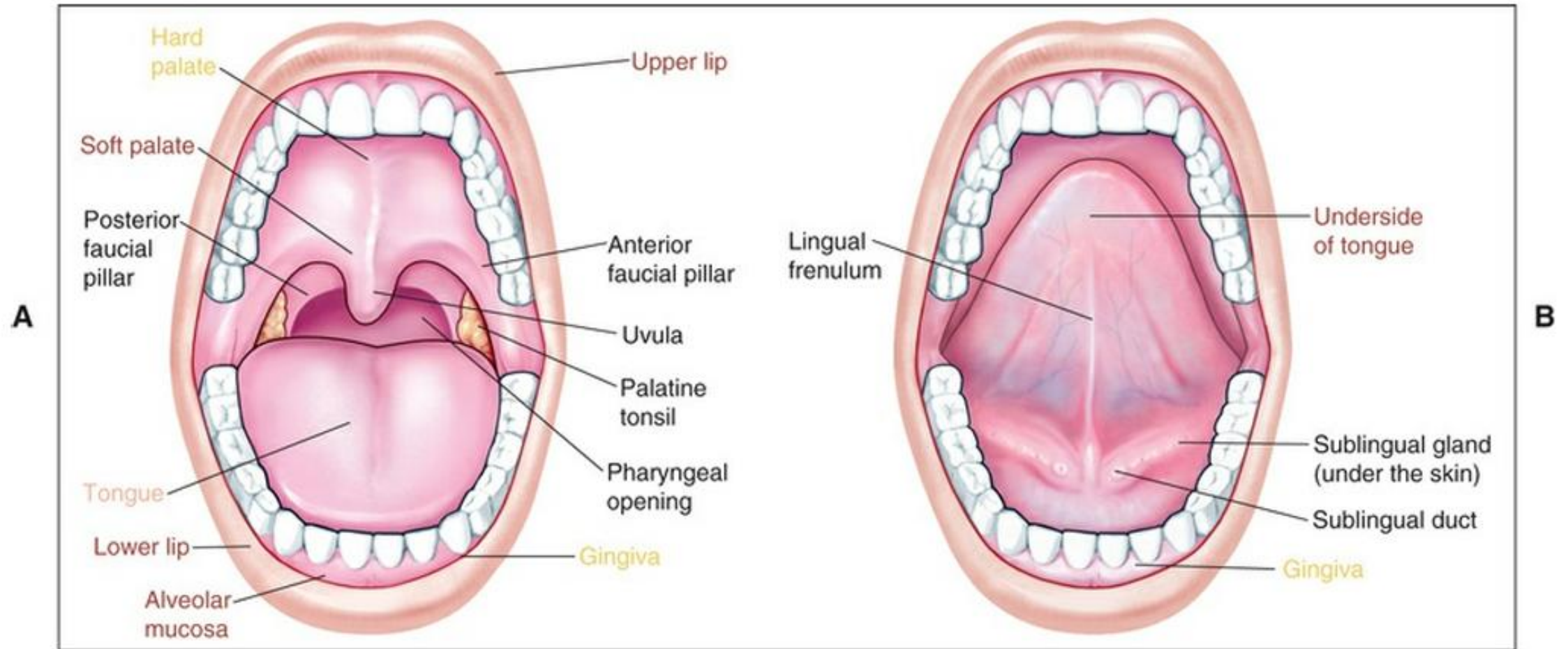
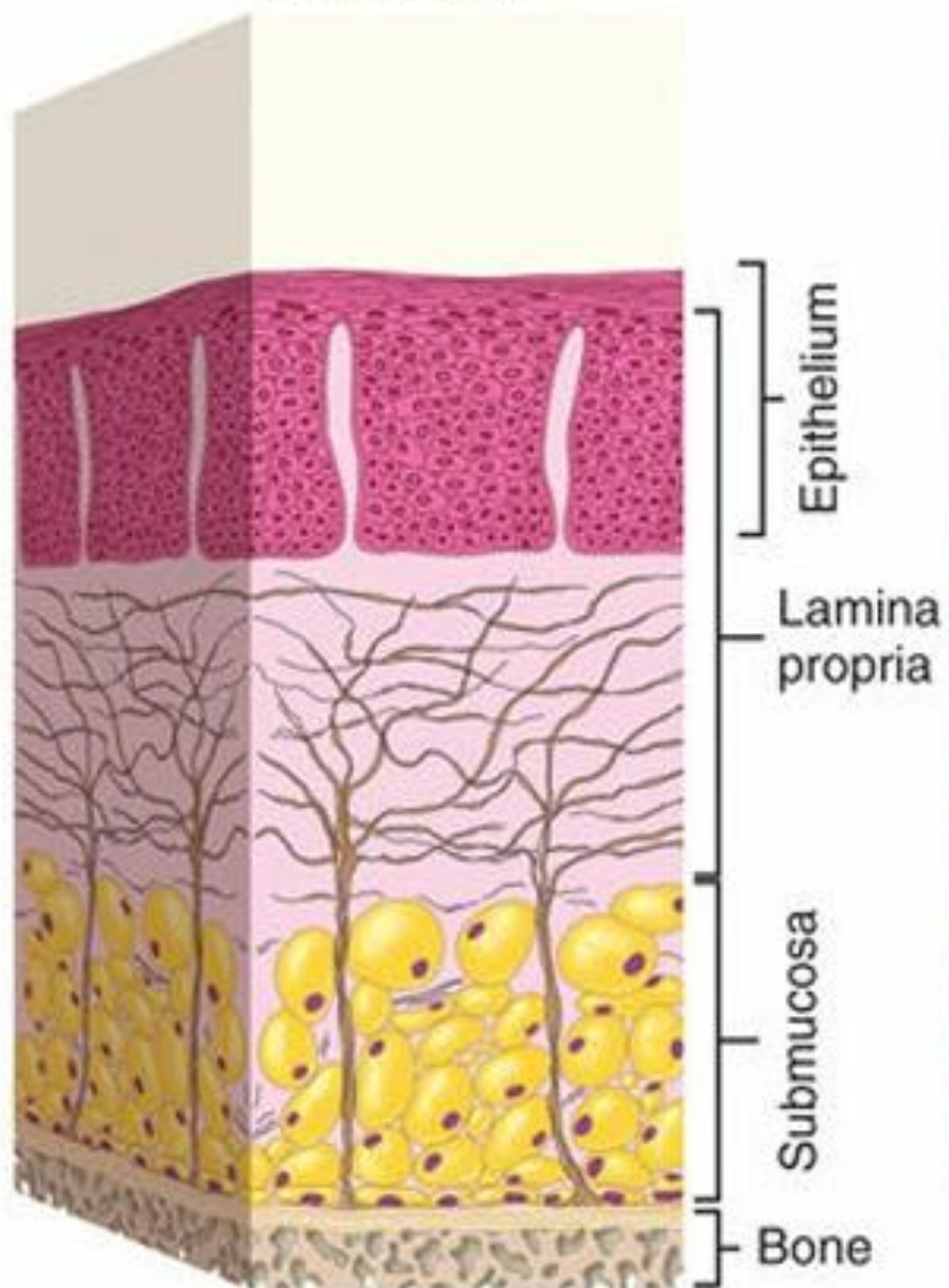
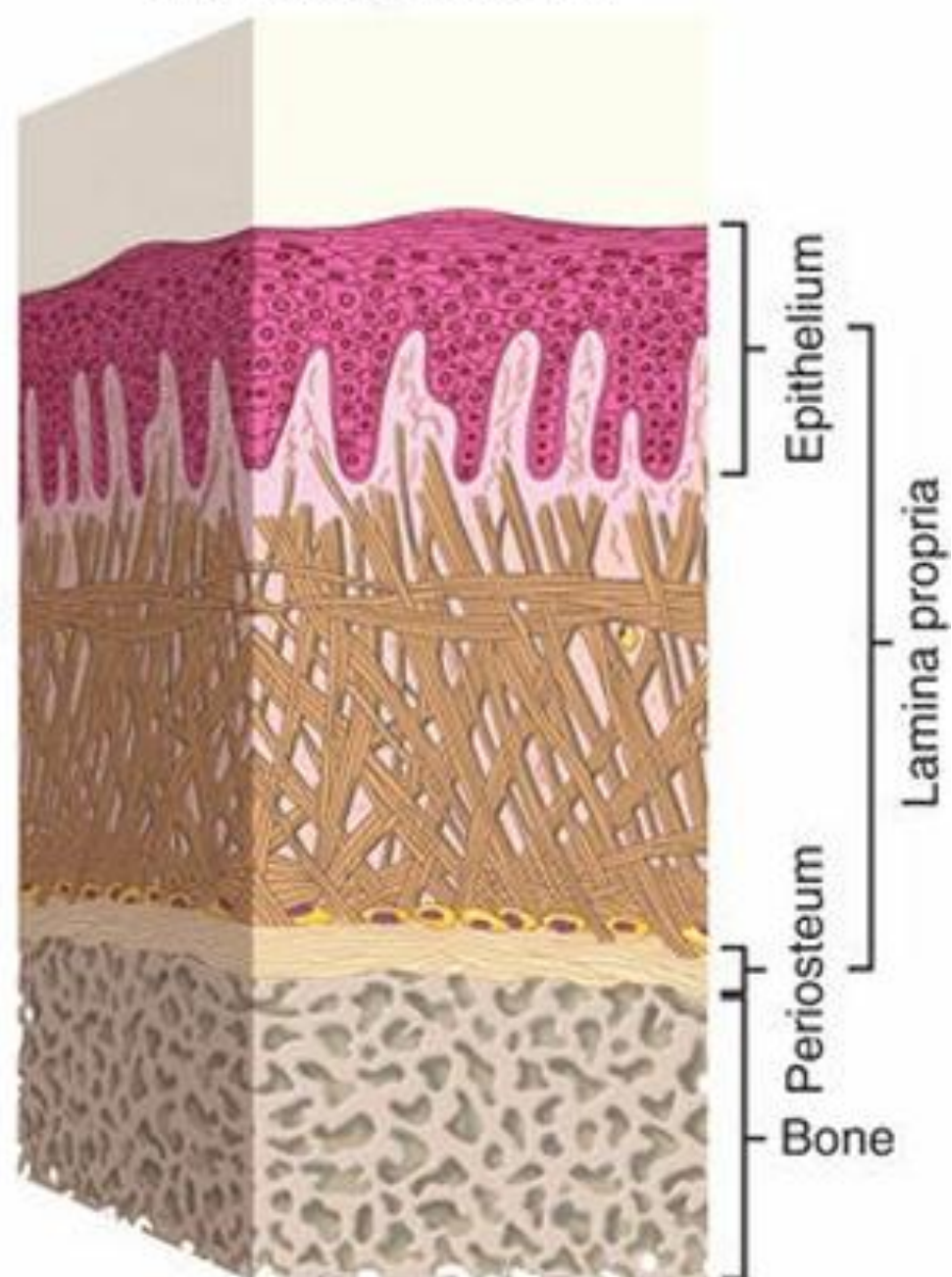


FIGURE 12-1 Anatomic locations occupied by the three main types of mucosa in the oral cavity. (From Thibodeau G, Patton K: *Anatomy and physiology*, ed 6, St Louis, 2007, Mosby.)

Oral mucosa



Oral mucoperiosteum



A



B



Sebaceous glands in the mucosa of the cheek.
A, Clinically, these appear as clusters of yellowish spots called Fordyce's granules.
B, Histologic section of a biopsy from this region. Note the presence of minor salivary glands in proximity to the sebaceous glands.
(Courtesy A. Kauzman.)

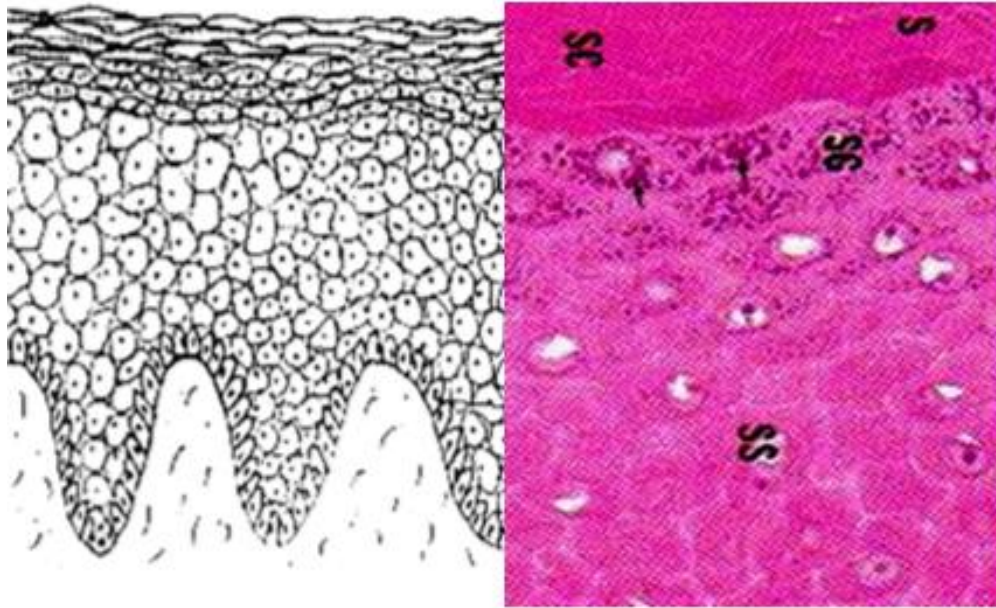
Structure of OMM

- The epithelium of the oral mucous membrane is of **the stratified squamous variety**.
- It may be **keratinized (orthokeratinized or parakeratinized) or nonkeratinized, depending on location**.
- Both the keratinized and the nonkeratinized epithelium consists of two groups of cells namely the **keratinocytes and the nonkeratinocytes**.

Keratinized epithelium

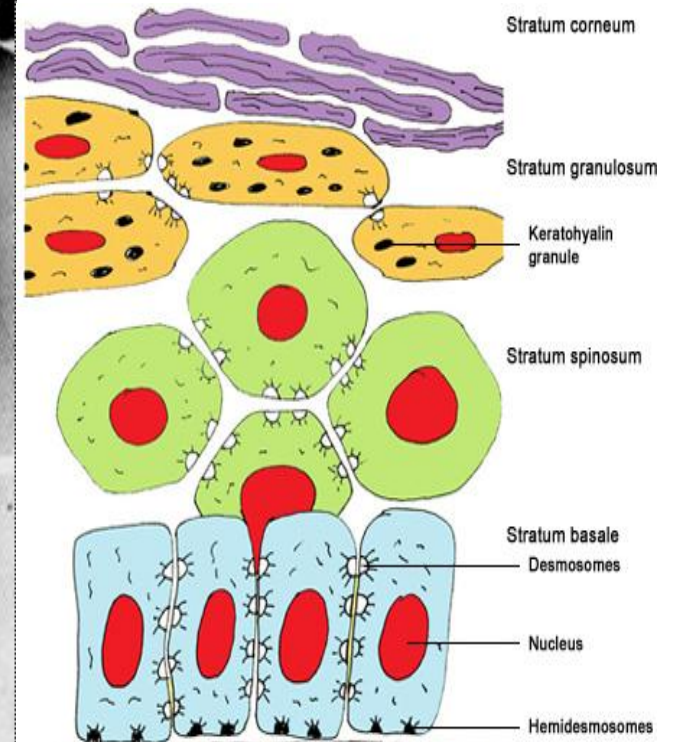
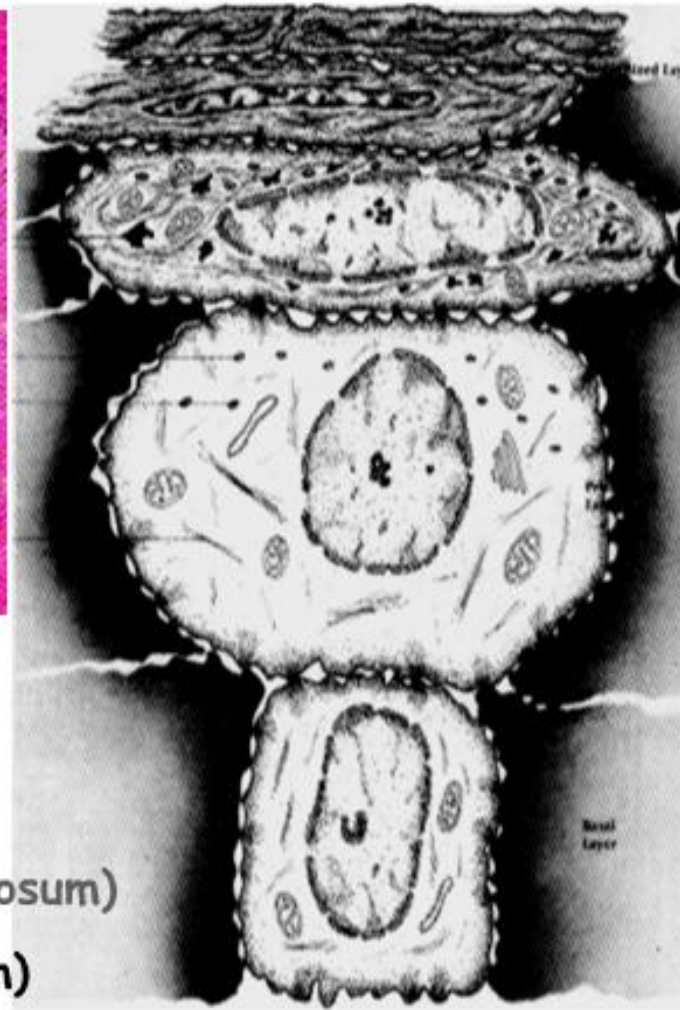
- Keratinizing oral epithelium has the **keratinocytes arranged in four cell layers: basal, spinous, granular, and cornified.**
- In Latin as
 1. Stratum basale,
 2. Stratum spinosum,
 3. Stratum granulosum,
 4. and stratum corneum

Components of keratinized stratified squamous epithelium



Keratinized St. Sq. Epith.

- Basal cell layer (stratum basale)
- Prickle 'spinous' cell layer (stratum spinosum)
- Granular cell layer (stratum granulosum)
- Cornified cell layer (stratum corneum)



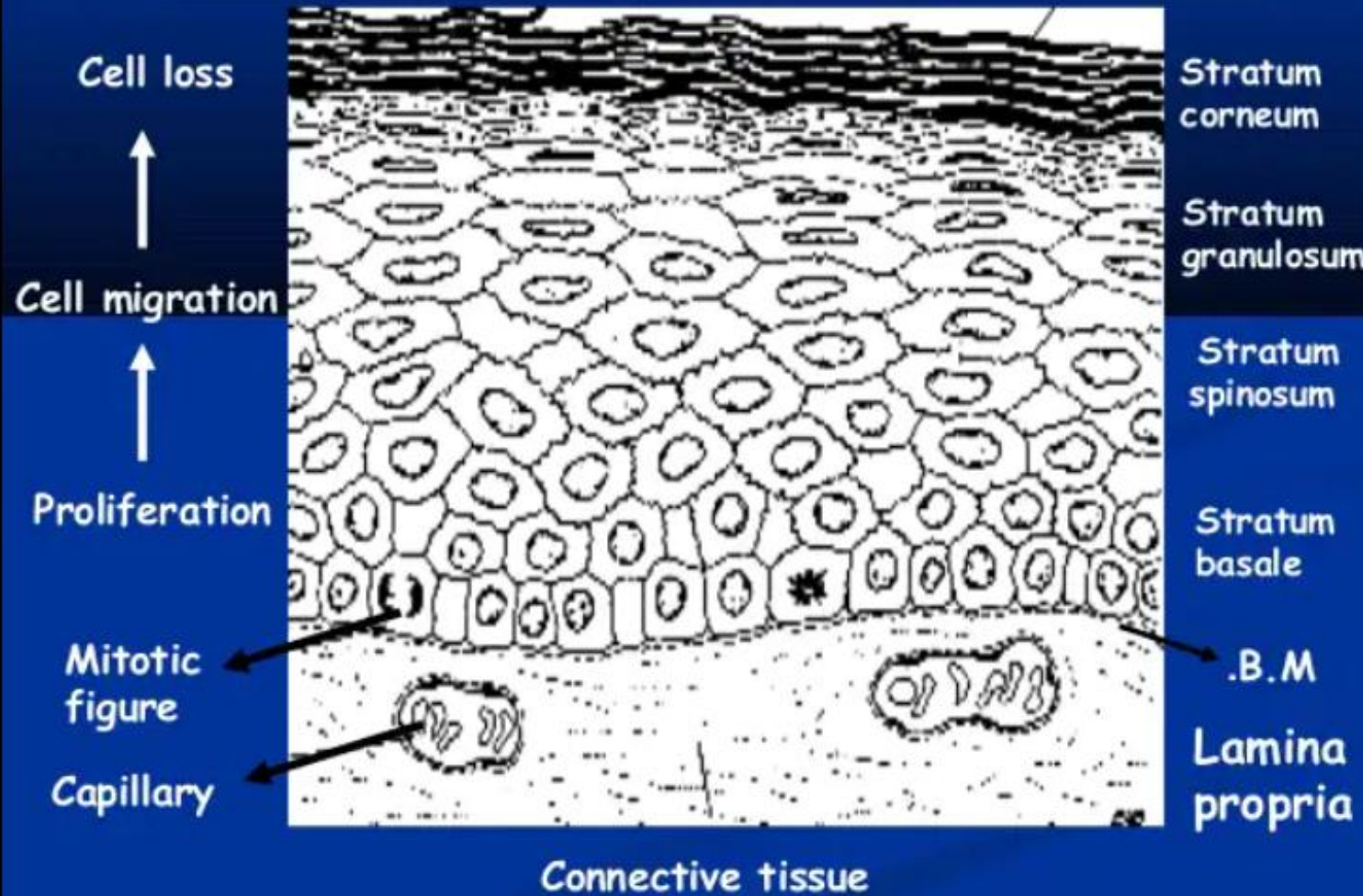
Differentiation

- **Mitosis** takes place in **basal cell layer**.
- After mitosis, Cells may remain in the basal layer and divide again or it may become determined, during which time it migrates and is pushed upward.
- **During its migration** as a keratinocyte it becomes committed to biochemical and morphologic changes. This is termed as **differentiation**.
- Differentiation ends with the formation of a keratinized squama, a dead cell filled with densely packed protein contained within a toughened cell membrane.

Desquamation, maturation and turnover time

- After reaching the surface it is shed or cast off. This process of shedding of surface epithelial cells is called **desquamation**.
- The process of cell migration from the basal layer to the surface is called **maturation**.
- The time taken for a cell to divide and pass through the entire epithelium is termed **turnover time or turnover rate**.
- The turnover time in **oral mucosa is faster** than skin. Turnover time of **nonkeratinized oral epithelium is faster** than keratinized oral epithelium.
- This whole process from the onset of determination is called keratinization.

Stratified Squamous Epithelium



Mitotic Division → New basal cells are produced.



Cell Migration Upward → Cells move passively toward the epithelial surface.



Maturation & Differentiation → Cells undergo progressive structural and functional changes.



Terminal Differentiation → Cells reach the surface, lose nuclei, and become flattened.

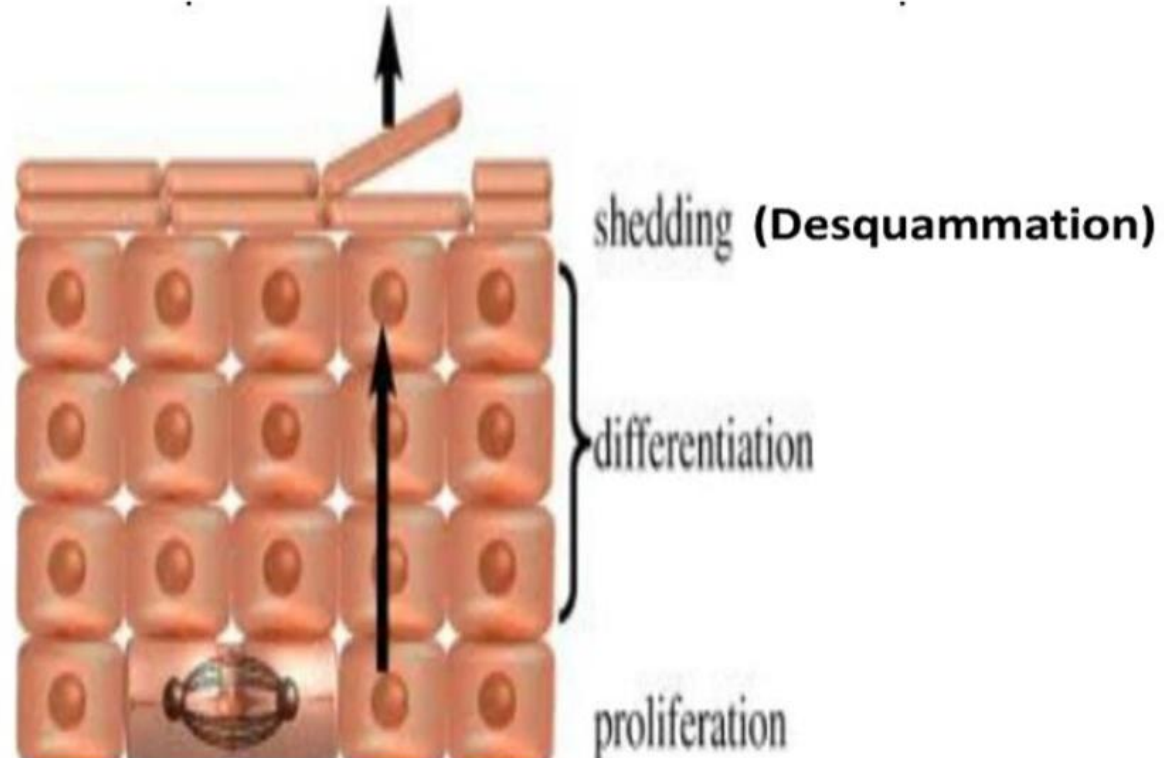
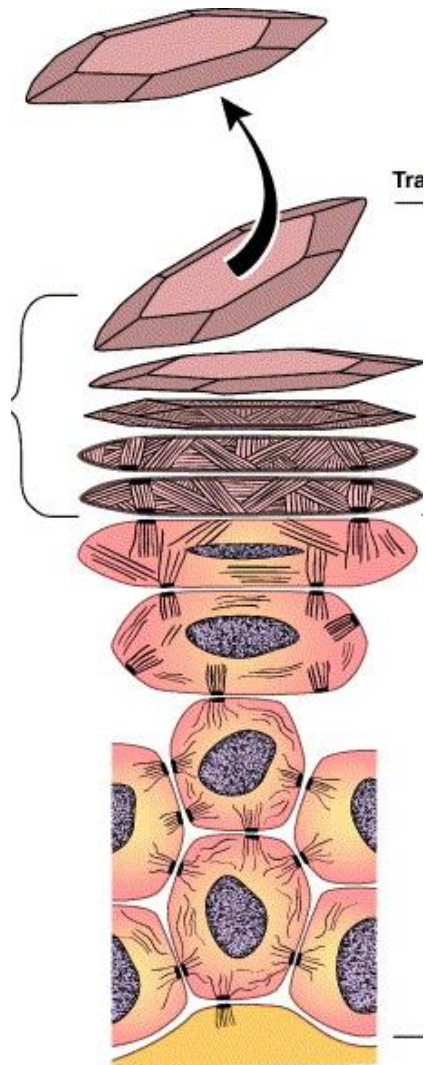


Shedding / Desquamation → Superficial cells are shed and replaced by newly formed ones

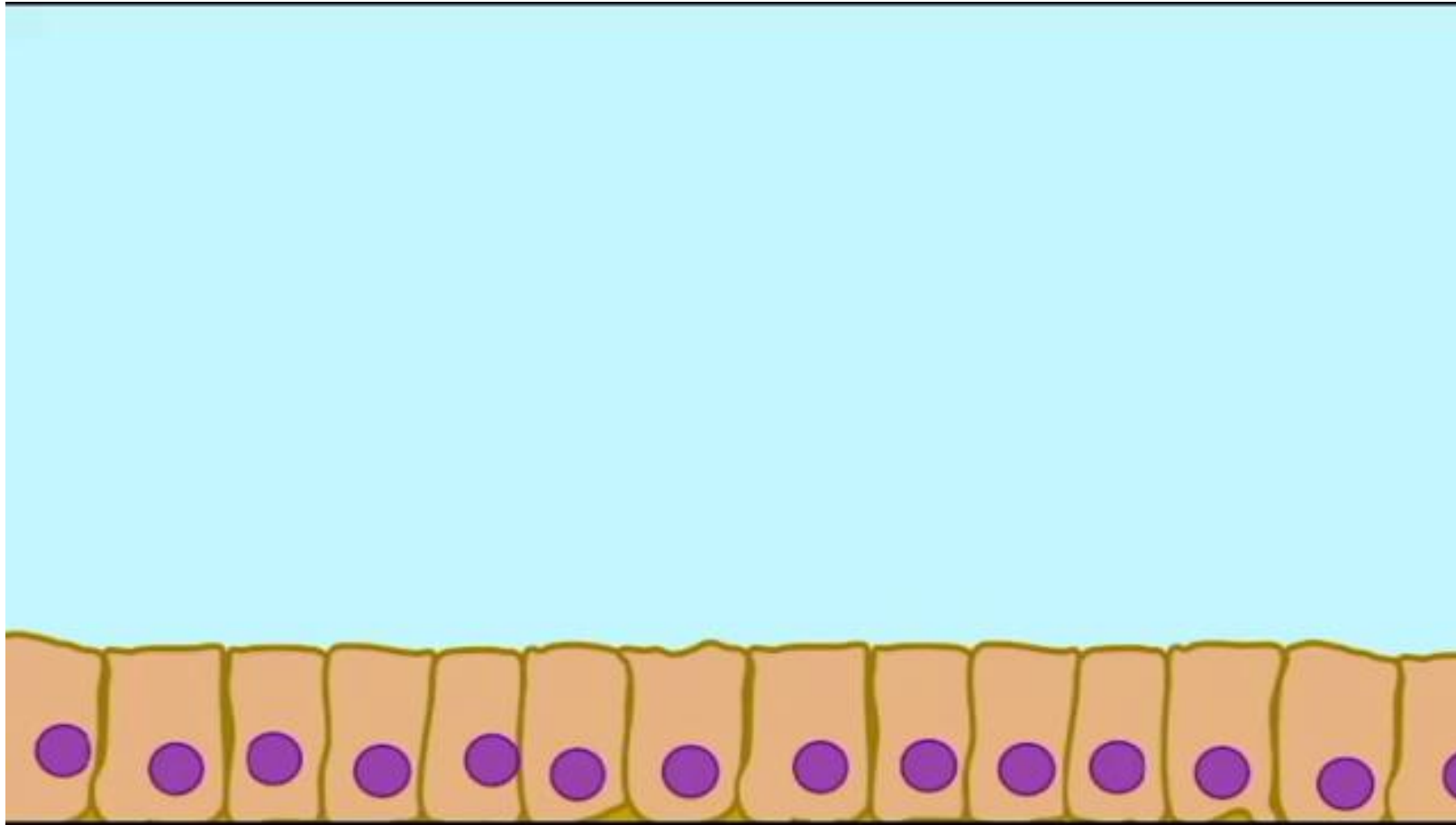
End Phase: Programmed Cell Death (Unique Form) → Final stage ensuring continuous renewal and tissue integrity.

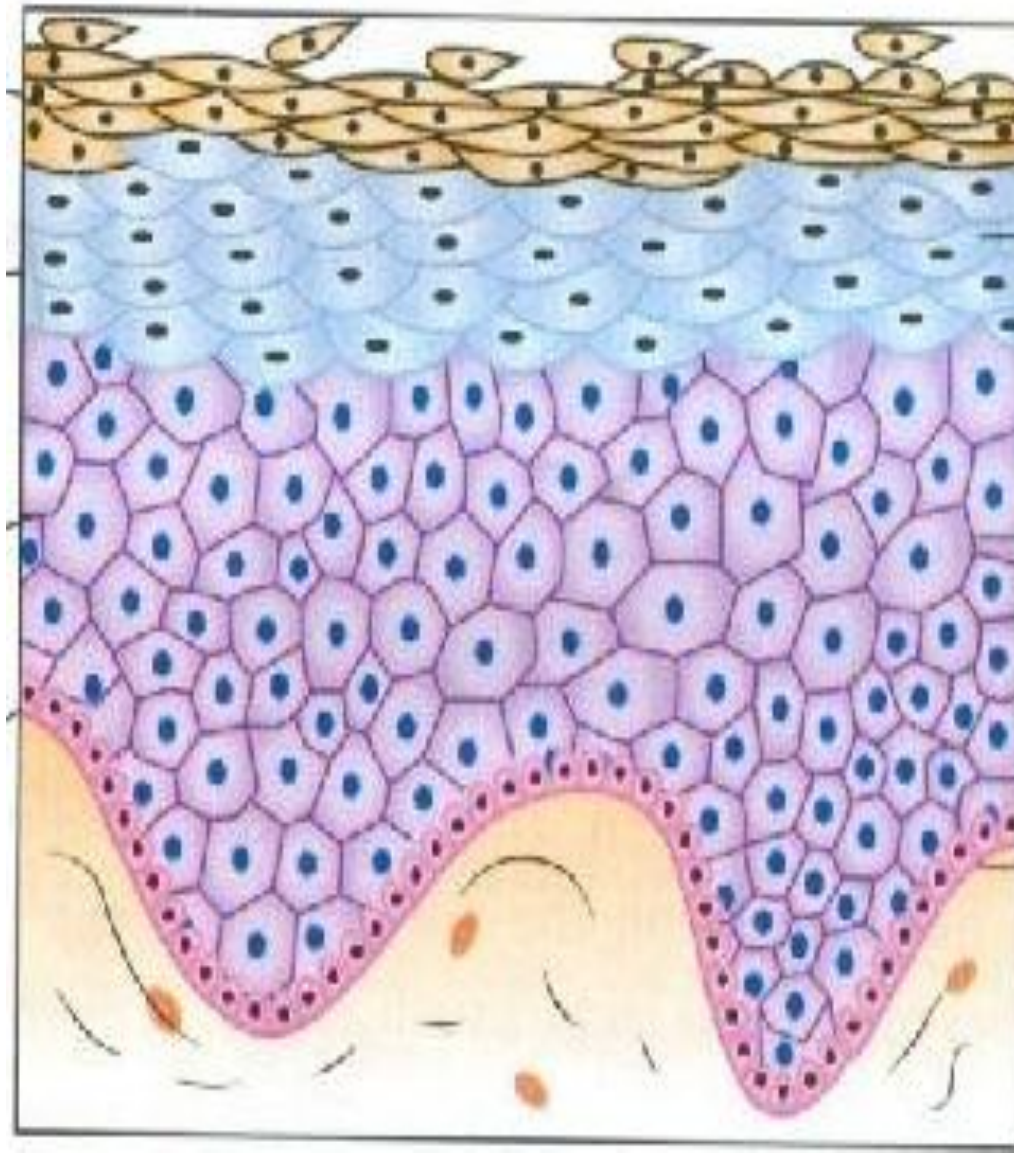
Epithelial cell turnover

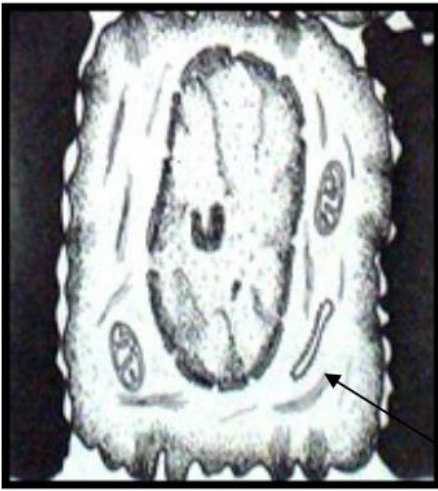
Replacement of old epithelial cells by new cells.



Shedding or desquamation of old cells at the superficial cell layer.







STRATUM BASAL (BASAL CELL LAYER)

The basal layer is made up of cells that synthesize DNA and **undergo mitosis**, thus providing new cells.
Also K/as stratum germinativum.



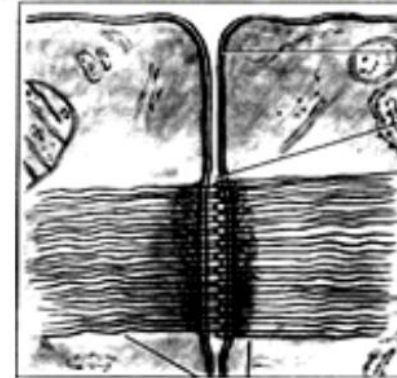
Keratinized .st.sq epith

Non-Keratinised.st.sq.epith

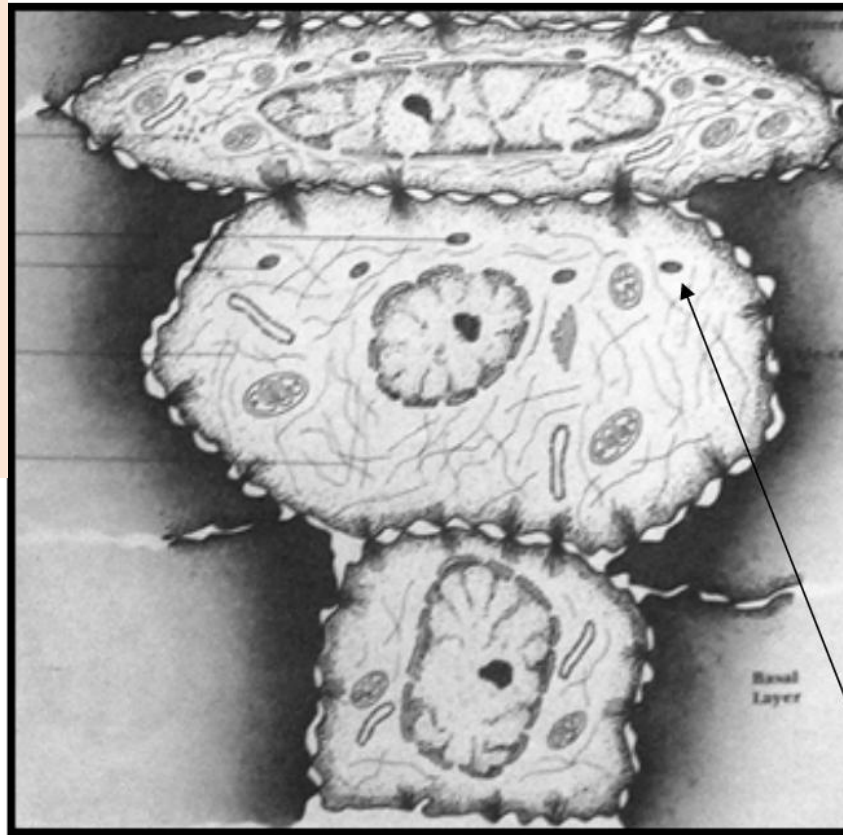
*The cells are cuboidal, contain large oval nucleus & basophilic cytoplasm

*The cytoplasm contain **tonofilaments**

*The cells are closely attached **to each other** by **desmosomes** & to **underlying basement membrane** by **hemidesmosomes**

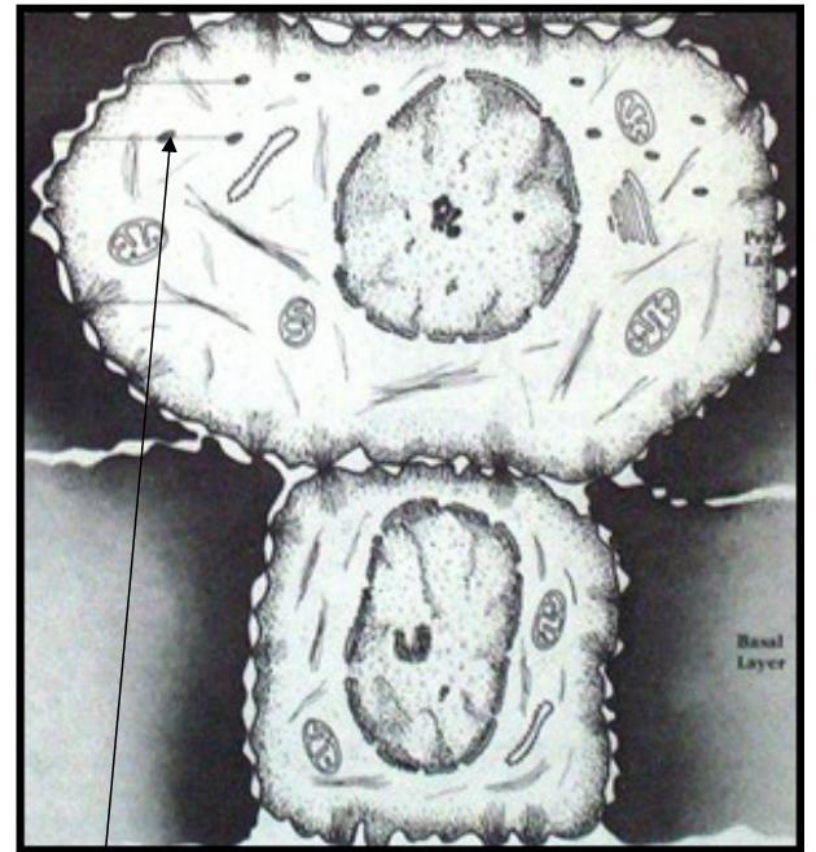


The spinous cells which make up this layer are irregularly polyhedral and larger than the basal cells. On the basis of light microscopy, it appears that the cells are joined by “intercellular bridges”



Keratinized St sq epith
(St. Spinosum)

- Cells are small, large I.C. spaces, long bridges
(Prickly app.)



Non-keratinised St sq epith
(St. Intermedium)

- Cells are large, small I.C. spaces
(No prickly app.)

Cytoplasm of superficial cells contain **Ondine Body (Keratinosome)**
The membrane coating granules are glycolipids.



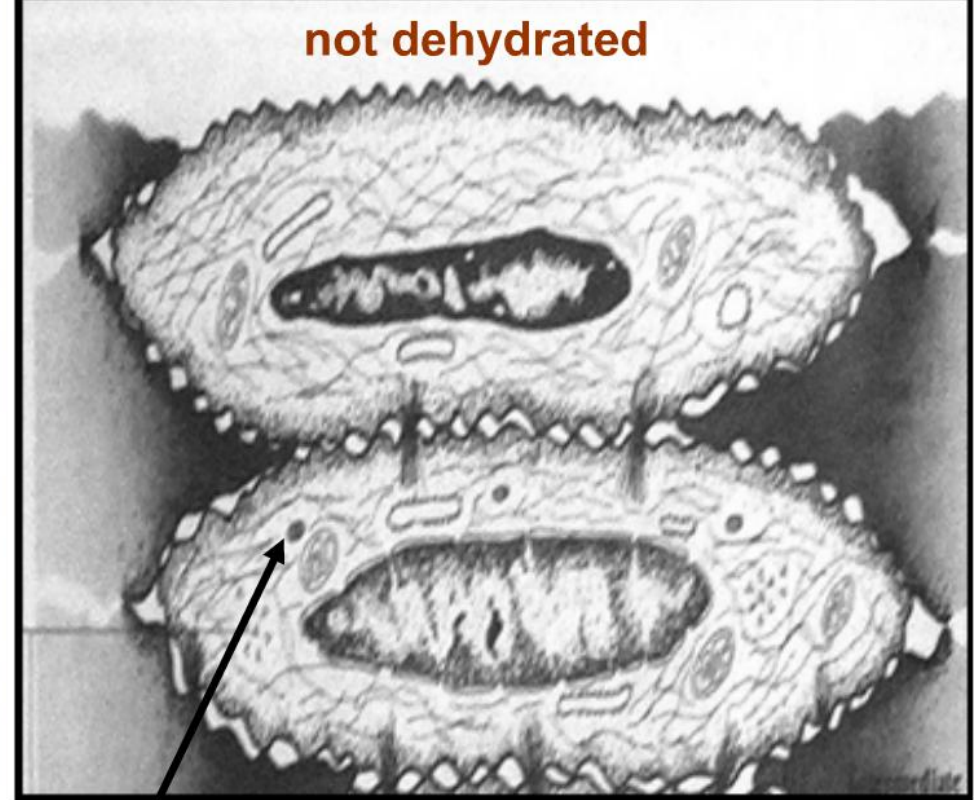
K.St sq epith (st. Granulosum)

Granules are called keratohyaline granules K.H.G

K.H.G

ass. with tonofilament bundles

Organelles decrease



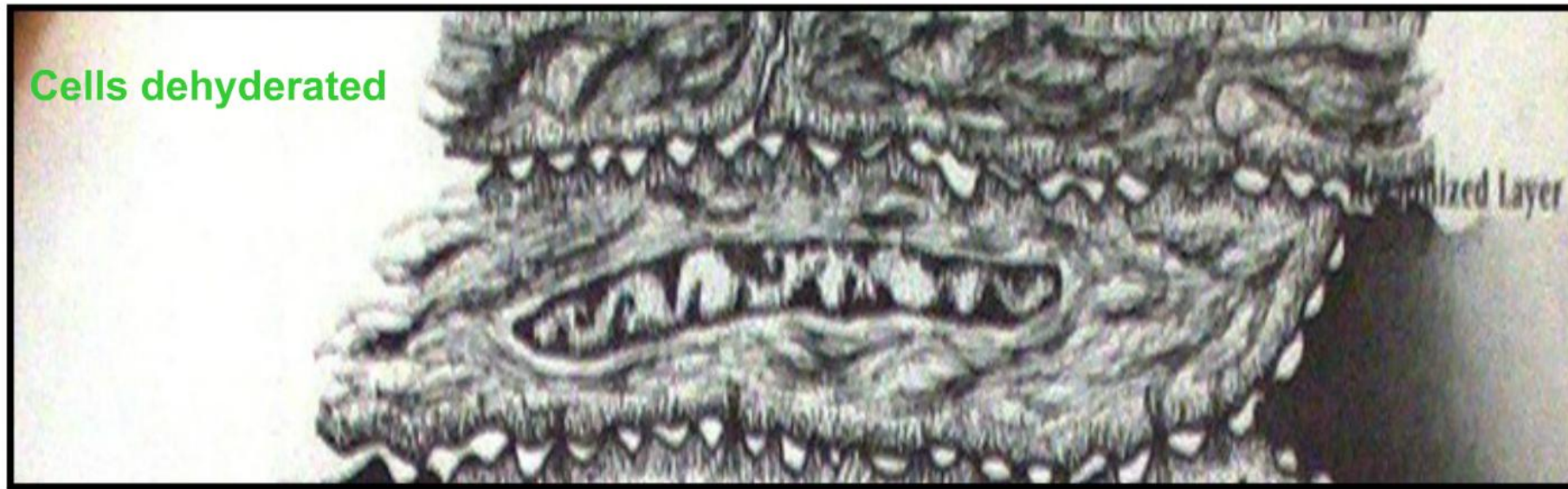
not dehydrated

Non-k st sq epith(st. Superficial)

K.H.G (rarely seen)

not ass.with tonofilament

Organelles decrease



Cornified layer (st. Cornium)

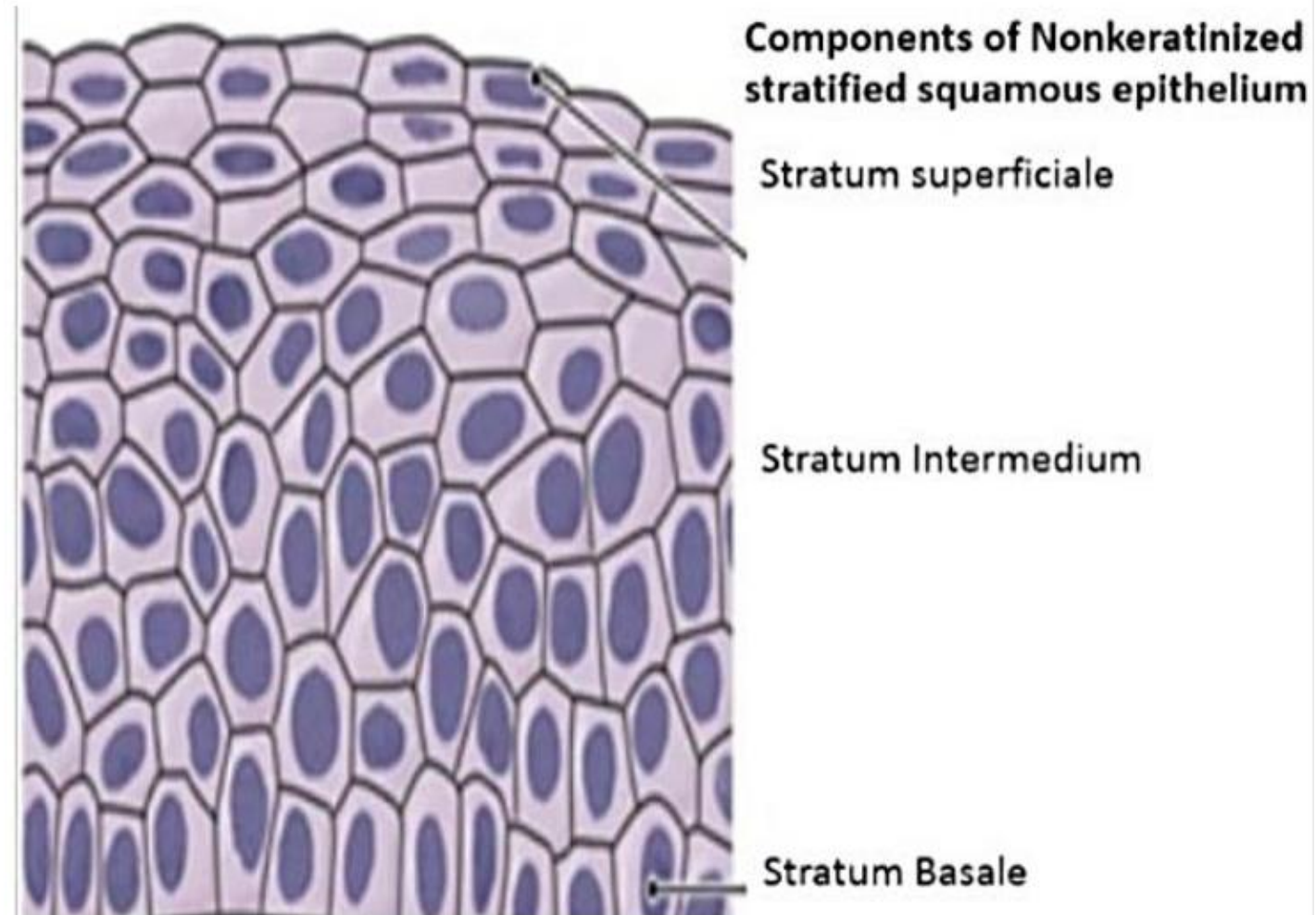
- Complete loss of **K.H.G** and **organelles**
- Cells filled by **tonofilaments** embedded in **fillagrin** (soft keratin) that replace the cytoplasm.
- **Desmosomes** weaken & disappear and are replaced by **surface interdigitations** to allow for **desquamation** and **they are always replaced by cells from the the underlying layers** The layer is acidophilic (dark pink staining with hematoxylin and eosin) and is histologically amorphous

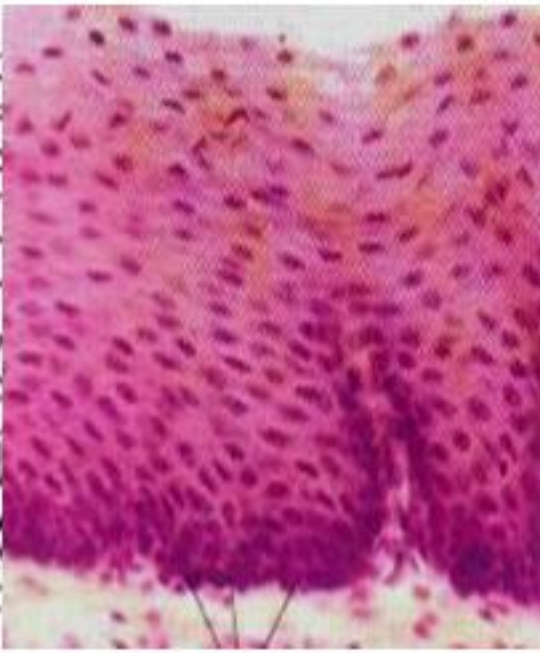
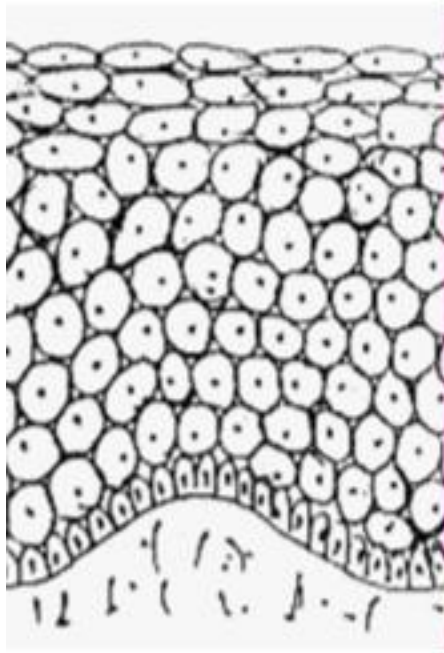
Components of Nonkeratinized Stratified Squamous Epithelium

1. STRATUM BASALE

2. STRATUM INTERMEDIUM

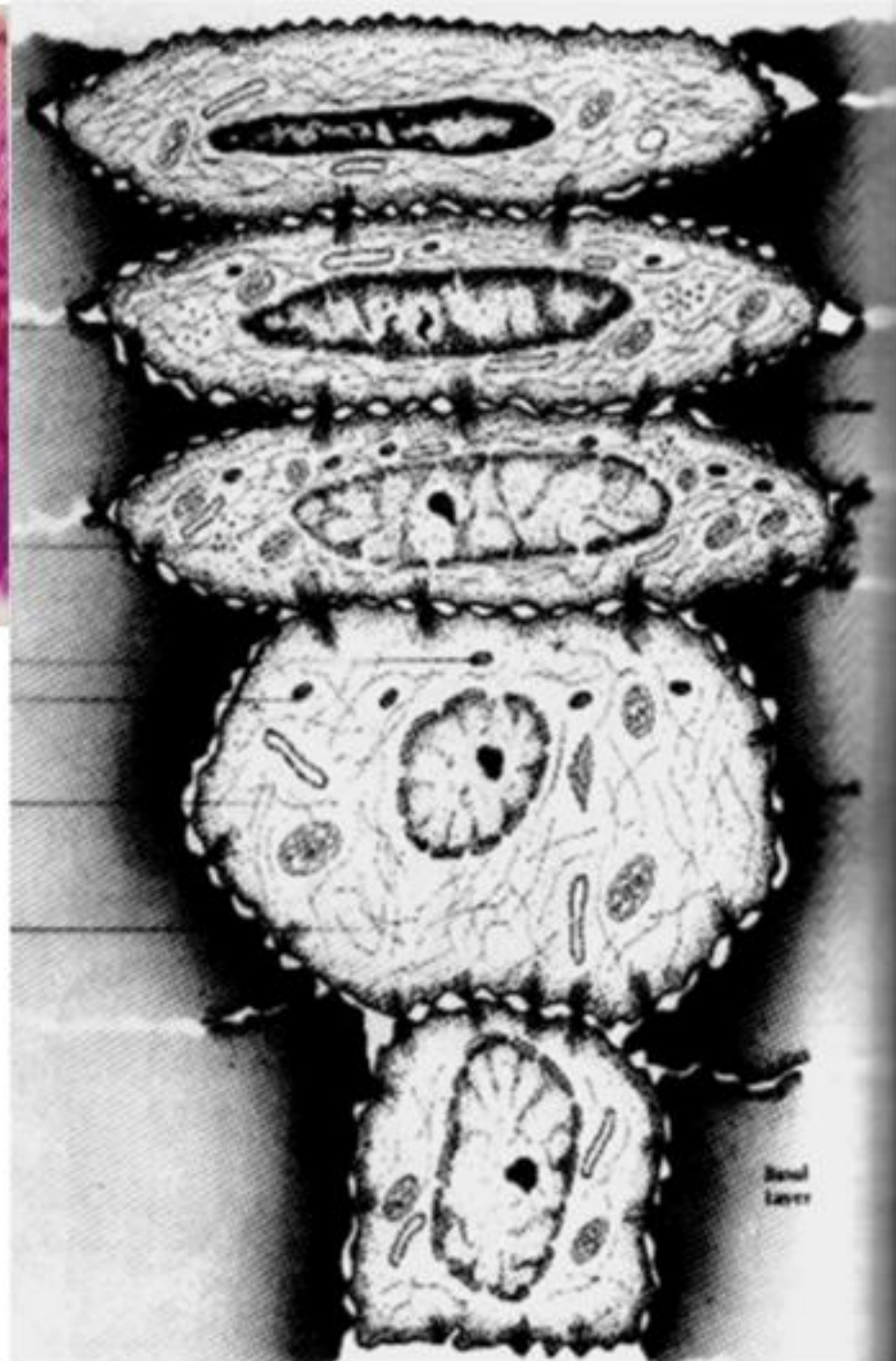
3. STRATUM SUPERFICIALE



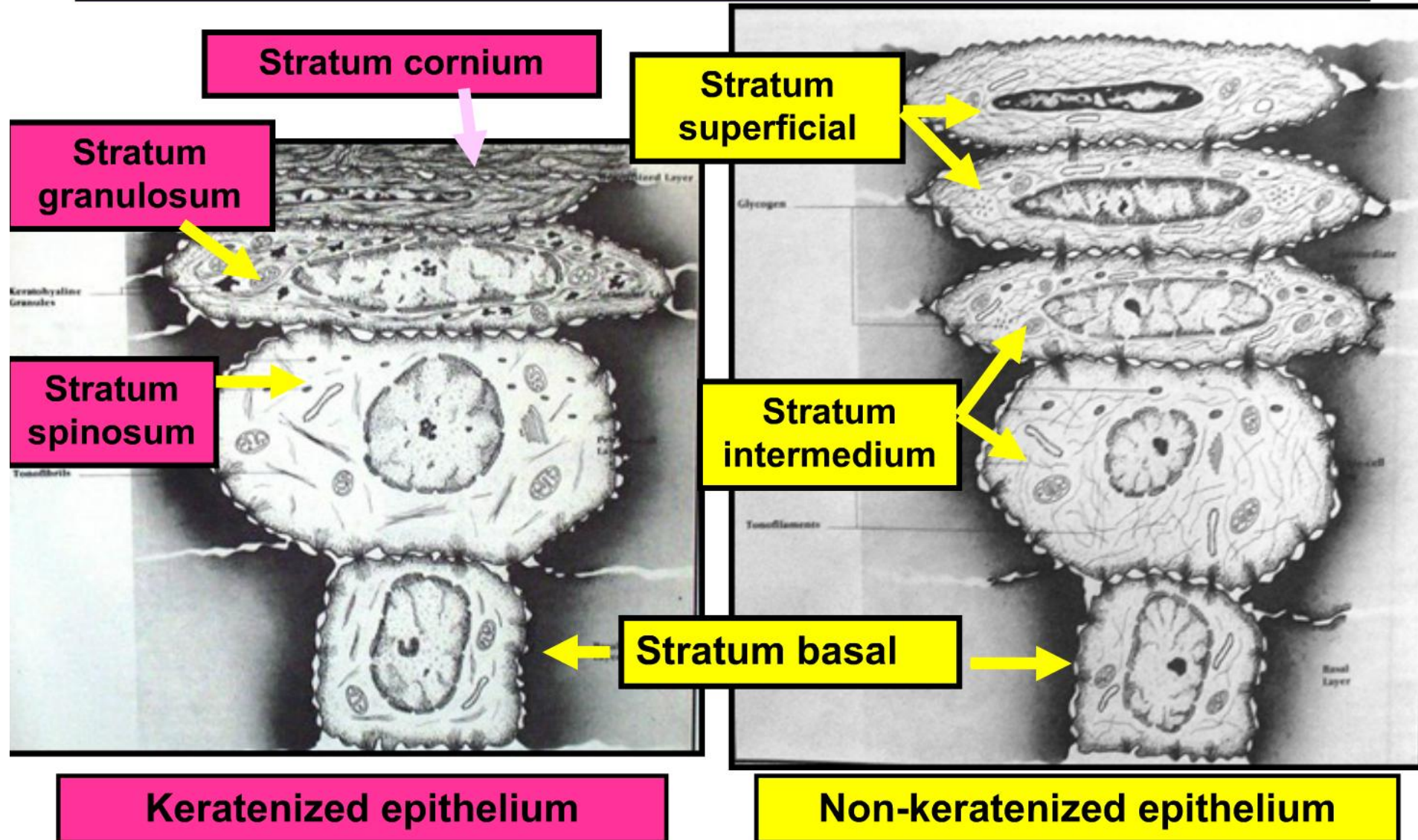


Nonkeratinized St. Sq. Epith.

- basal
- intermediate
- superficial

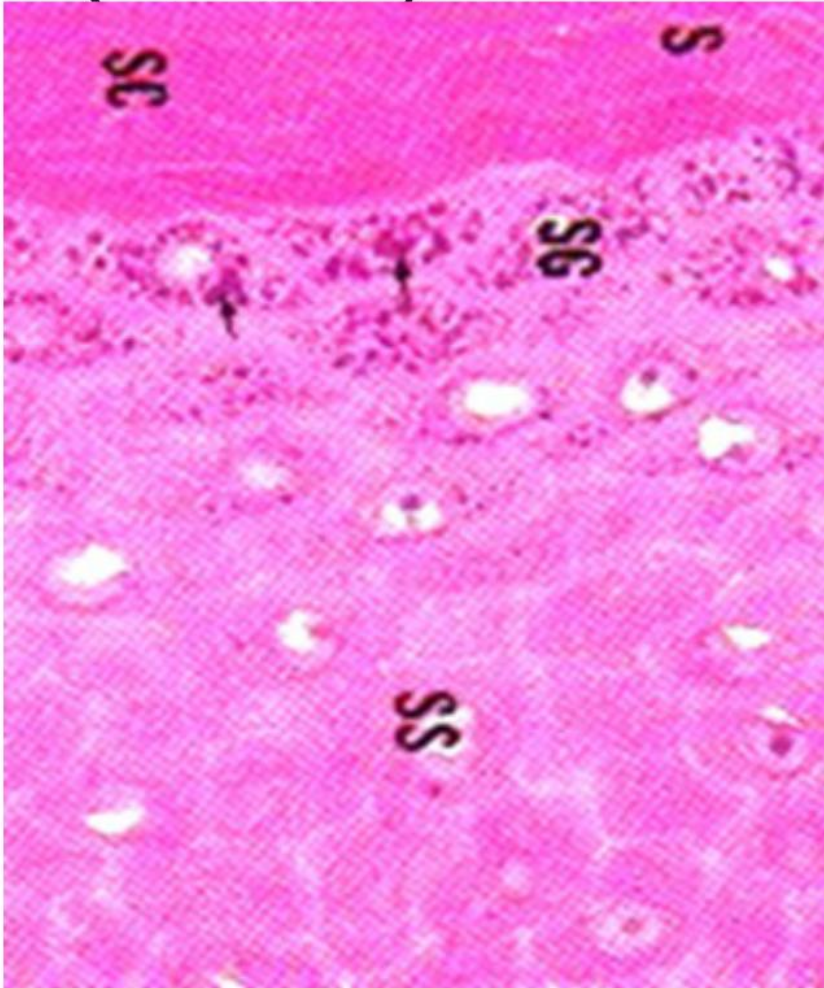


Keratinized and non-keratinized mucosa

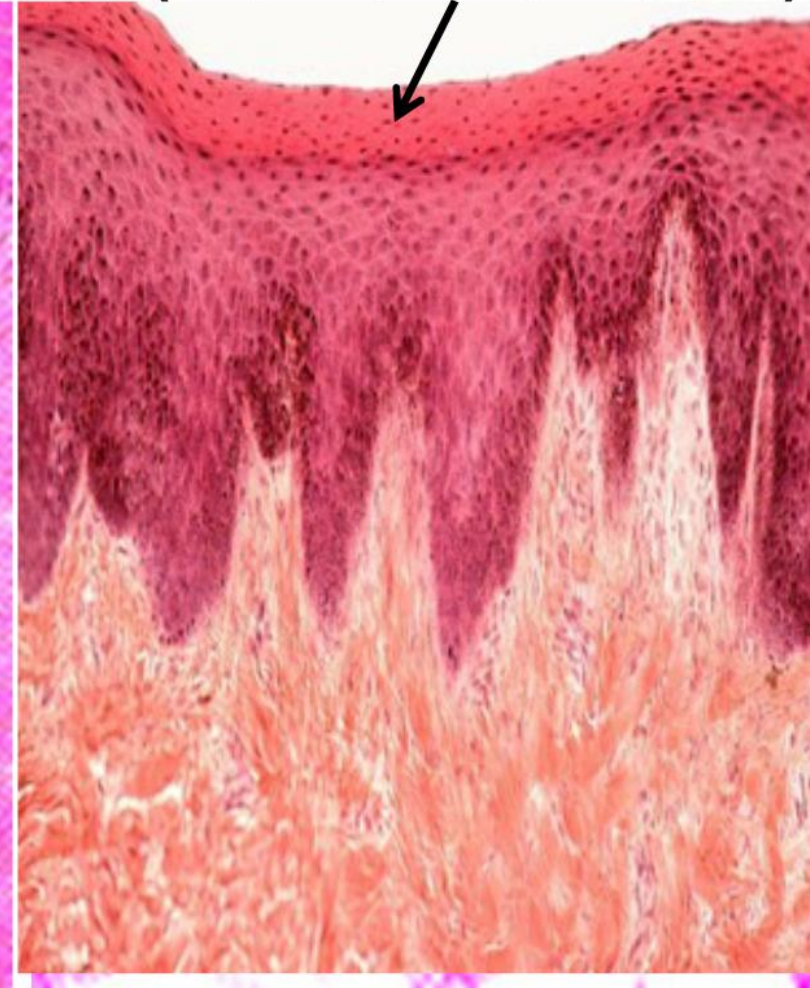


Types of Keratinization

orthokeratinized or
(No Nuclei)



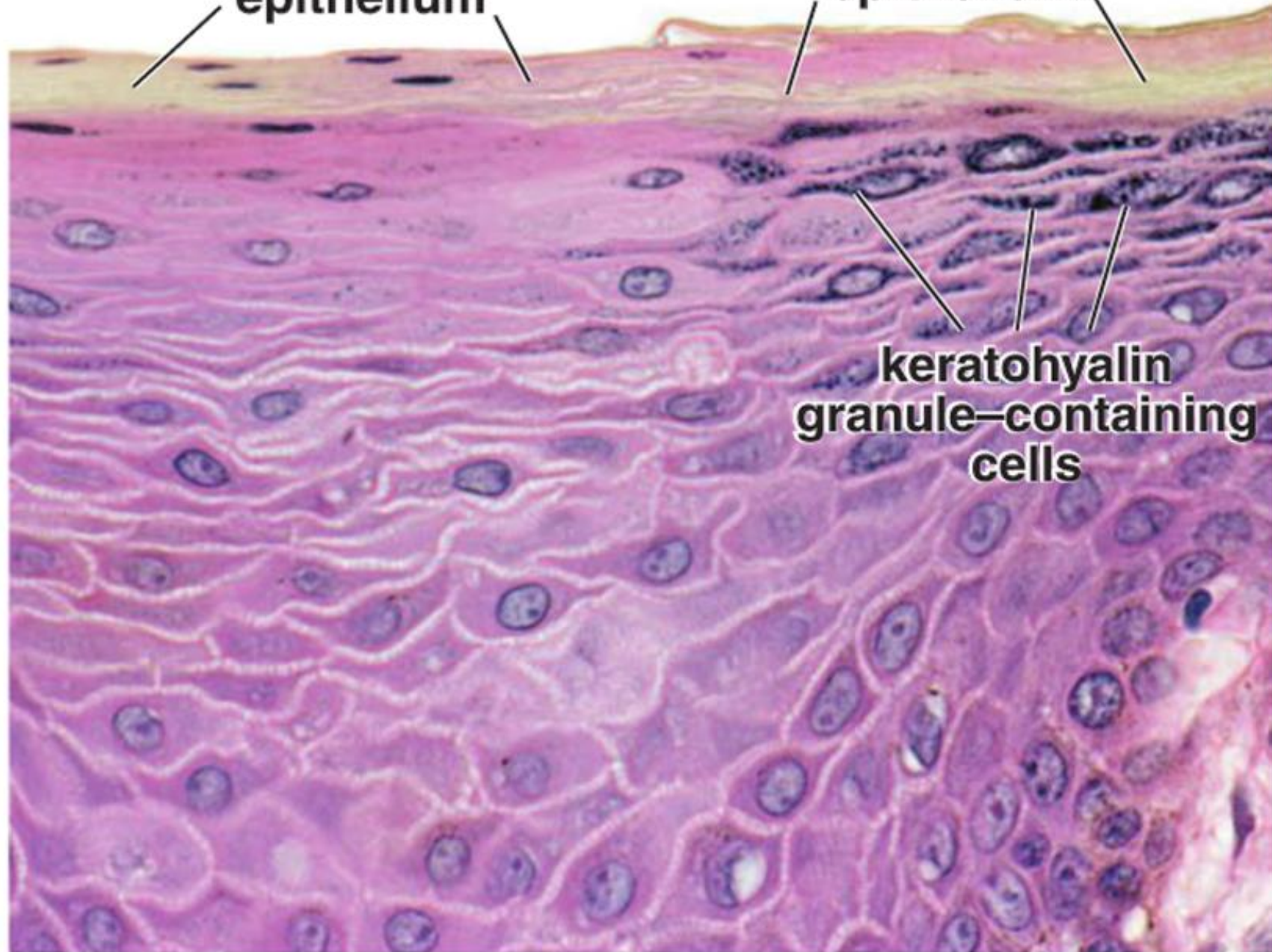
parakeratinized
(Remnants of nuclei)



**parakeratinized
epithelium**

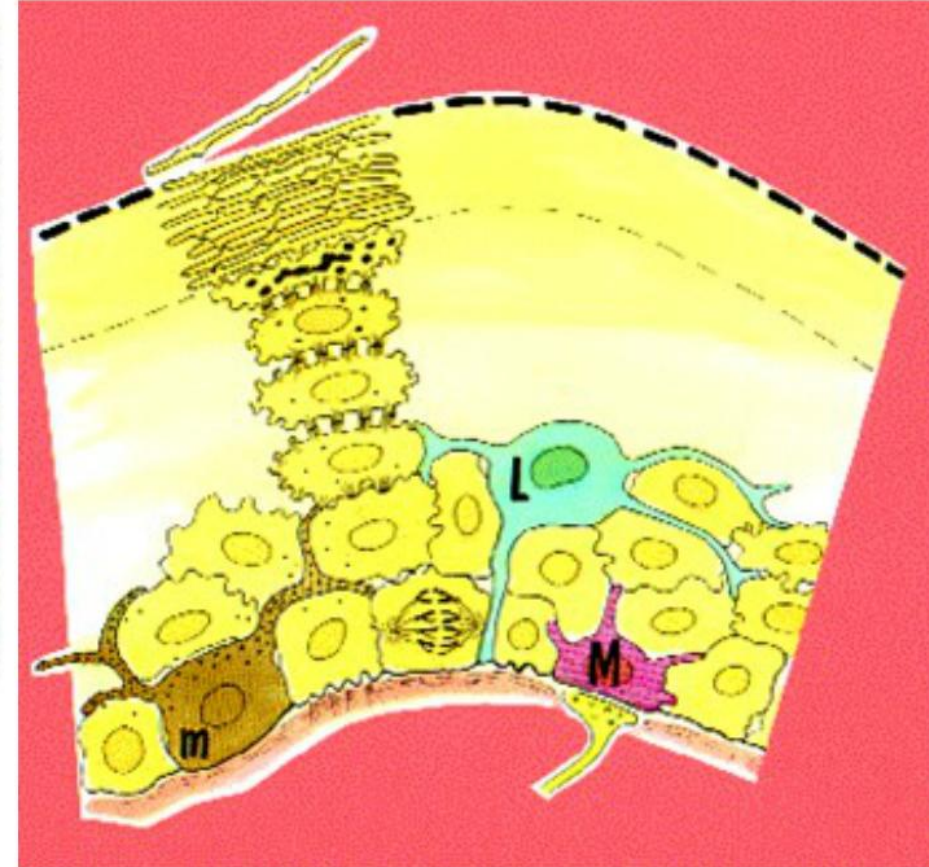
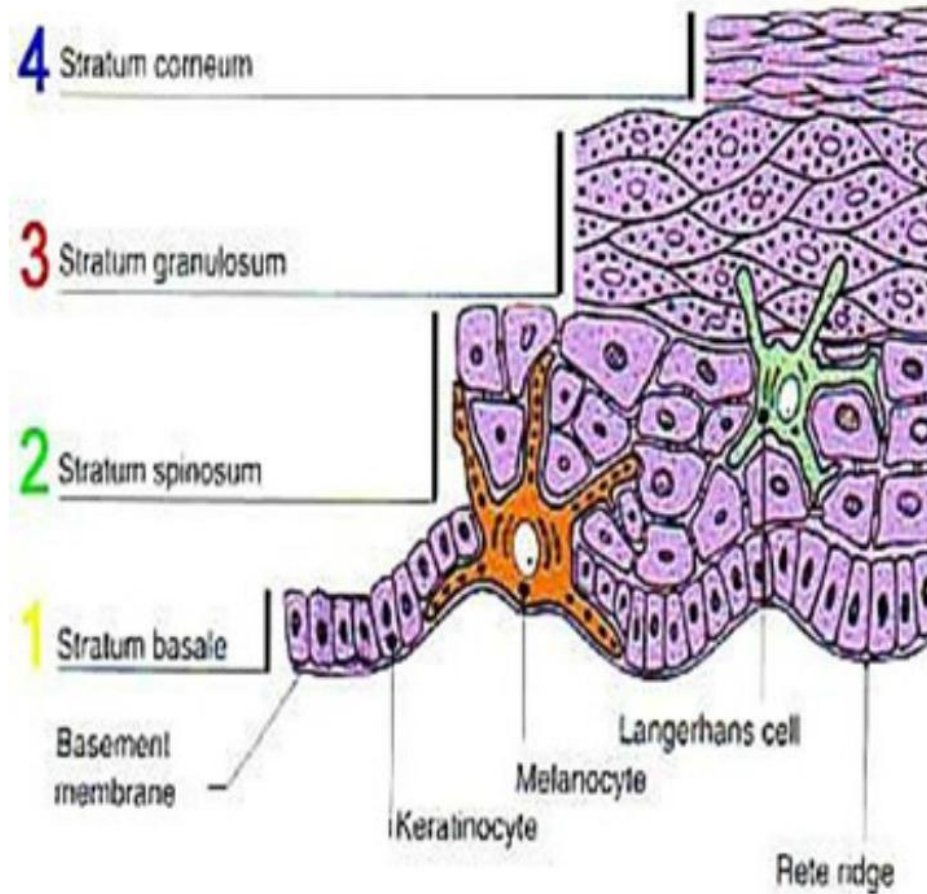
**keratinized
epithelium**

**keratohyalin
granule-containing
cells**



TYPES OF EPITHELIAL CELLS

Keratinocytes(90%) and non-keratinocytes(10%)



TYPE OF EPITHELIAL CELLS

Keratinocytes: 90% of epithelial cell population

Nonkeratinocytes: 10% of epithelial cell population

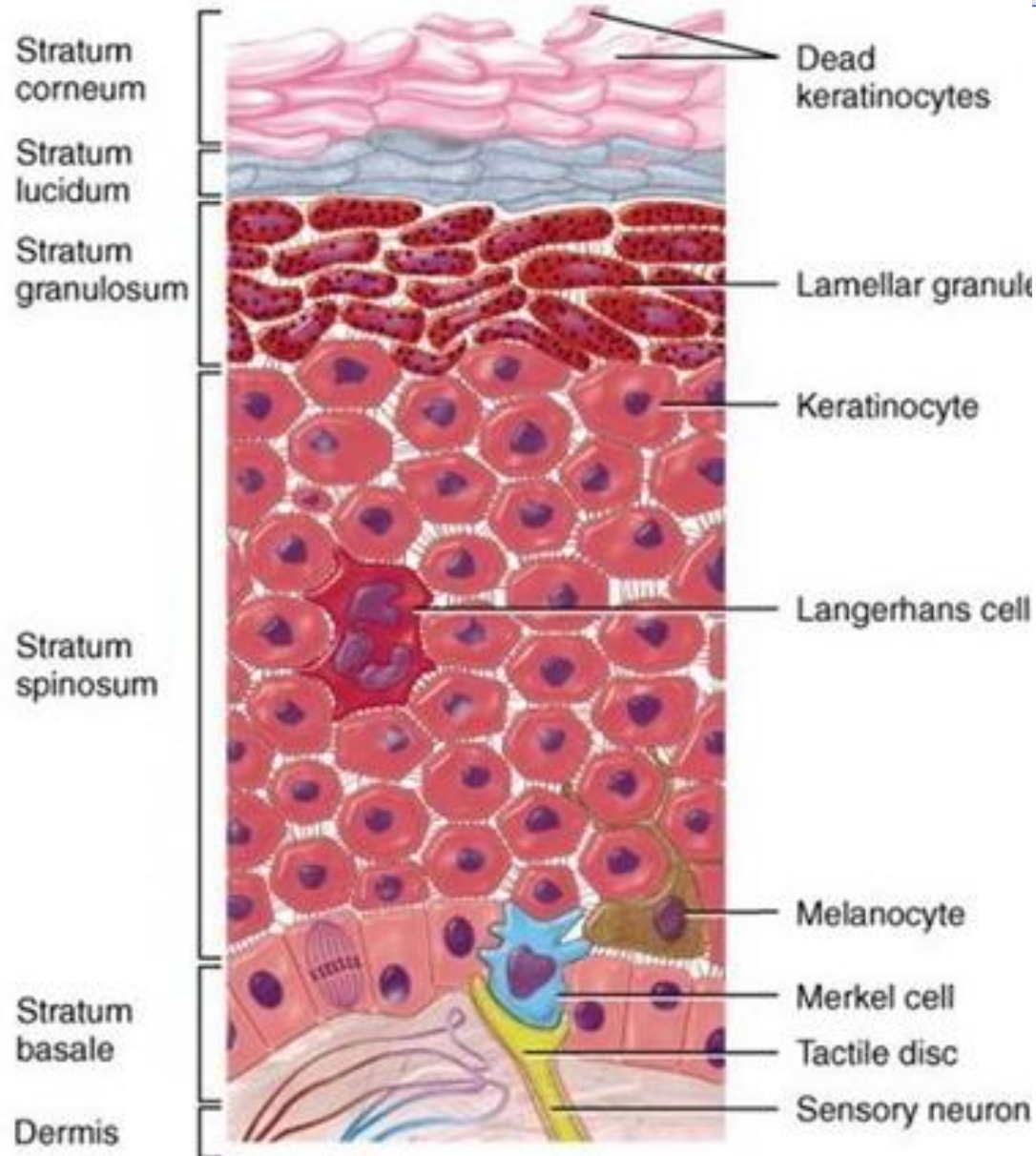
Langerhans cells: present in Stratum Spinosum (function is antigen trapping and processing)

Melanocytes: present in basal cell layer (contains melanin pigment) – mostly in gingiva.

Merkel cells: present in basal cell layer (act as touch receptor)

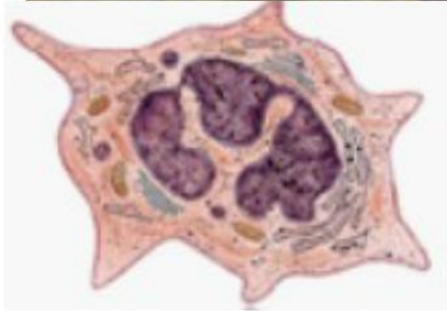
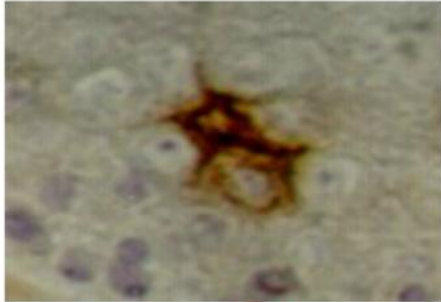
Lymphocytes: present in inflammation

Dendritic cells: Langerhans cells, melanocytes and Merkel cells are called as Dendritic cells due to the presence of cytoplasmic projections like dendrites. These cells appear clear cells in hematoxyline and eosin stain

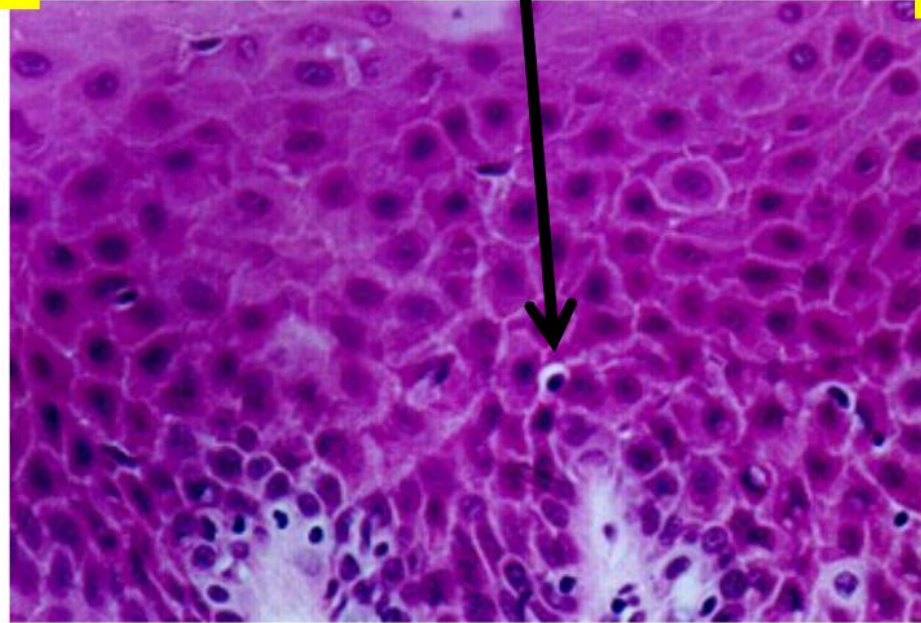


Nonkeratinocytes

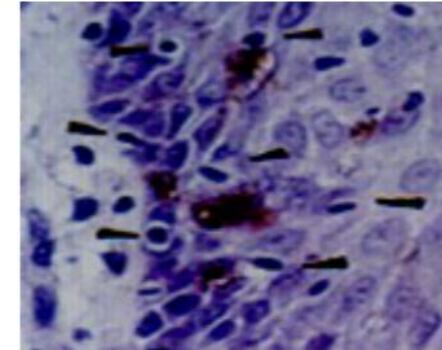
1-Langerhans Cells



Clear cells

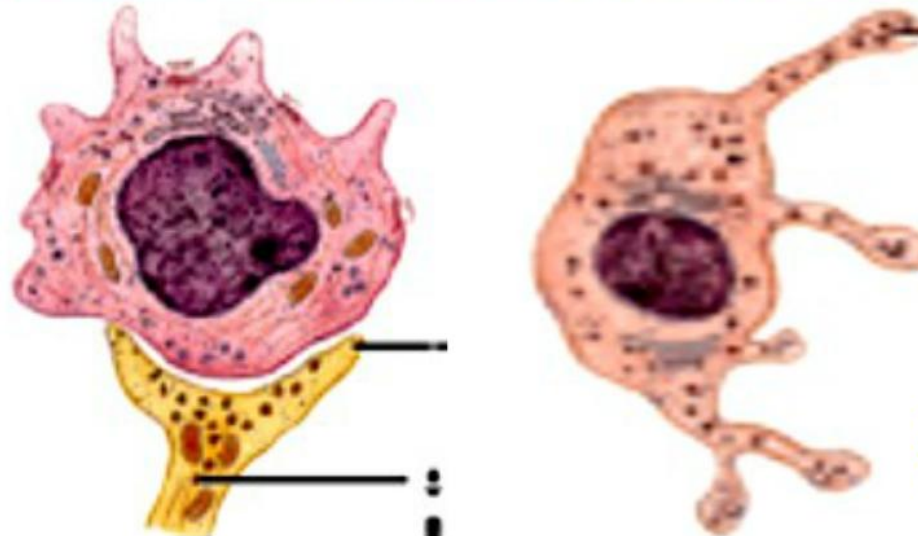


4- Lymphocytes & PMNL

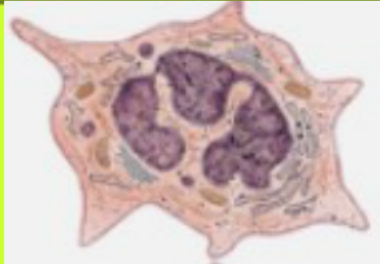


appear in
inflammation
only

2-Merkle cells



3- Melanocytes

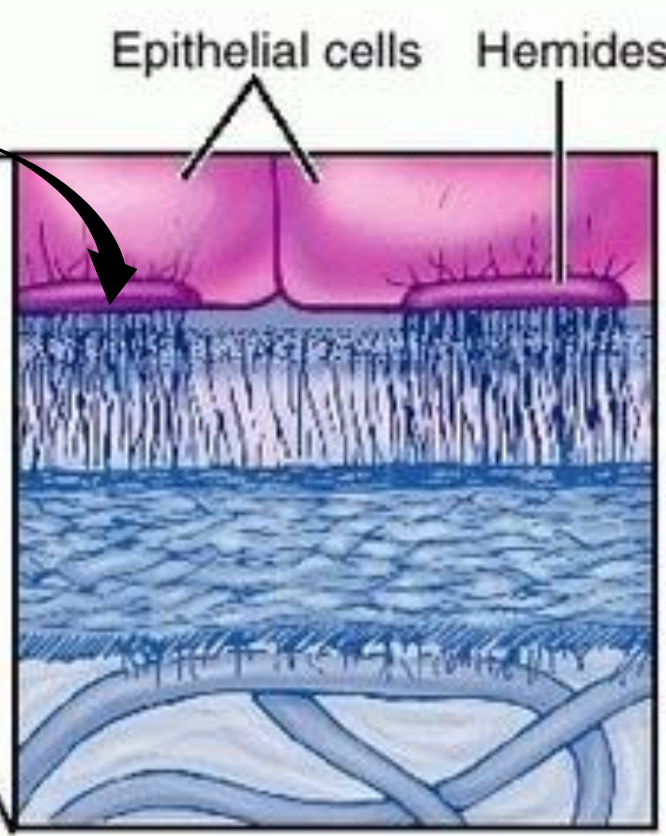
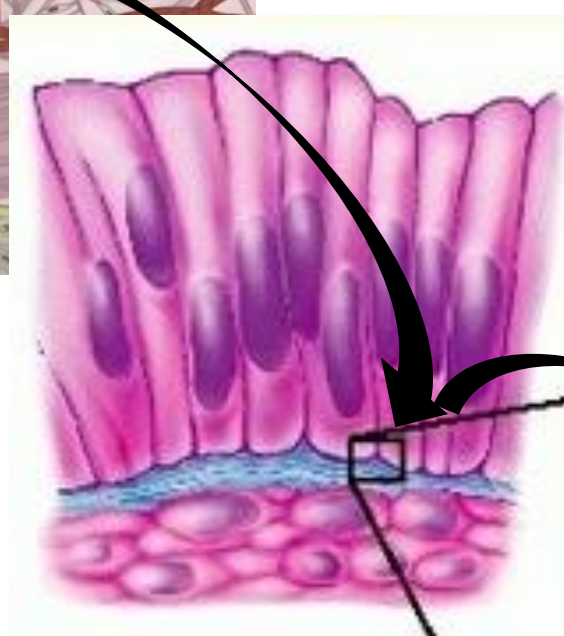
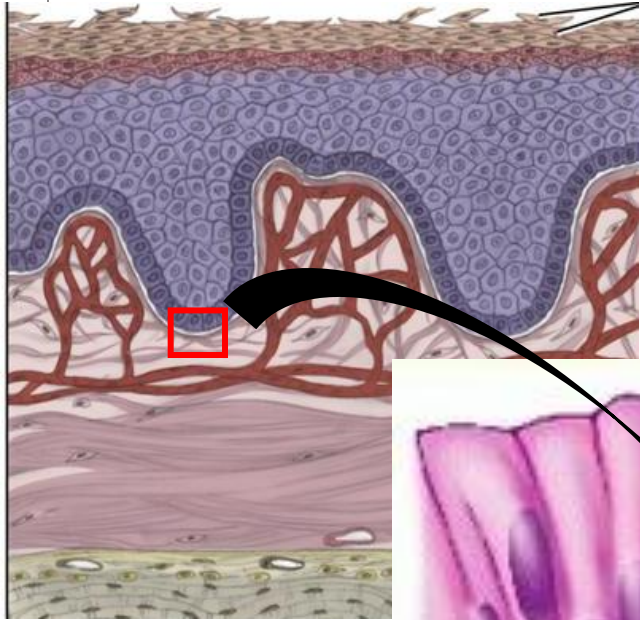
	1- Pigment cell (Melanocyte, blast)	2- Langerhan's cell	3- Merkel's cell
Shape	 -Have branched process -Present in the inter cellular spaces of epith. - contain melanin granules (melanosomes)	 - Similar in shape. - Contain rod like granules (Langerhan's granules) (Bir-beck granules)	 They do not have long processes. Contain small (membrane bounded granules)
Location	Basal and parabasal layers	High level cell	Basally located
Stain by H&E	Not stained so called (Clear dentritic cell)	Not stained so called (Clear dentritic cell)	Not stained so called(Clear but not dentritic cell)

- DIFFERENTIATION :
- DESQUMATION:
- MATURATION:
- TURNOVER TIME:



BASEMENT MEMBRANE (BM)

- The interface between the connective tissue and the epithelium in light microscopy appears thick.
- **1 to 4 μm wide** and is **relatively cell free**.
- It contains **neutral mucopolysaccharides (glycosaminoglycans)**. This zone stains positively with the **periodic acid-Schiff method**.
- **Ultra structurally,** BM is called **basal lamina**, The basal lamina is made up of **a clear zone (lamina lucida)** just below the epithelial cells and **a dark zone (lamina densa)** beyond the lamina lucida and adjacent to the CT.
- Basement membrane contains Protein **Laminin** and **type IV collagen**.

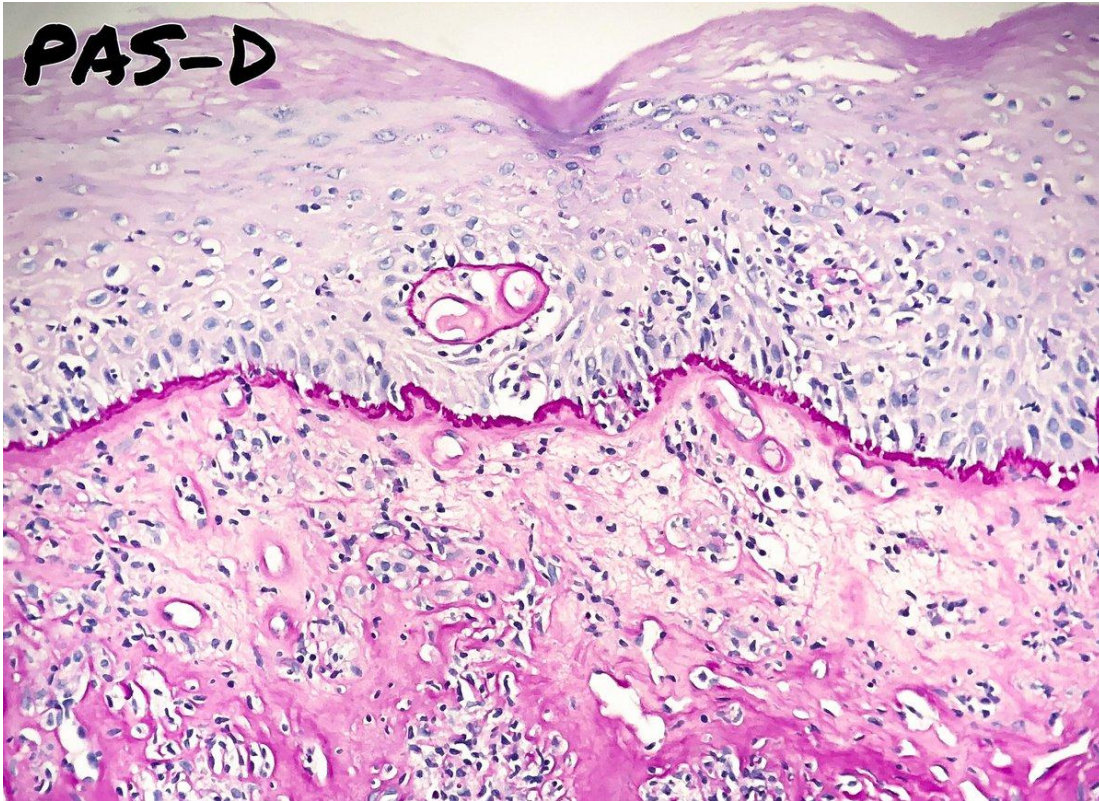


UNDER ELECTRON MICROSCOPE
(Ultrastructure of basal lamina.)

UNDER LIGHT MICROSCOPE
Basement membrane

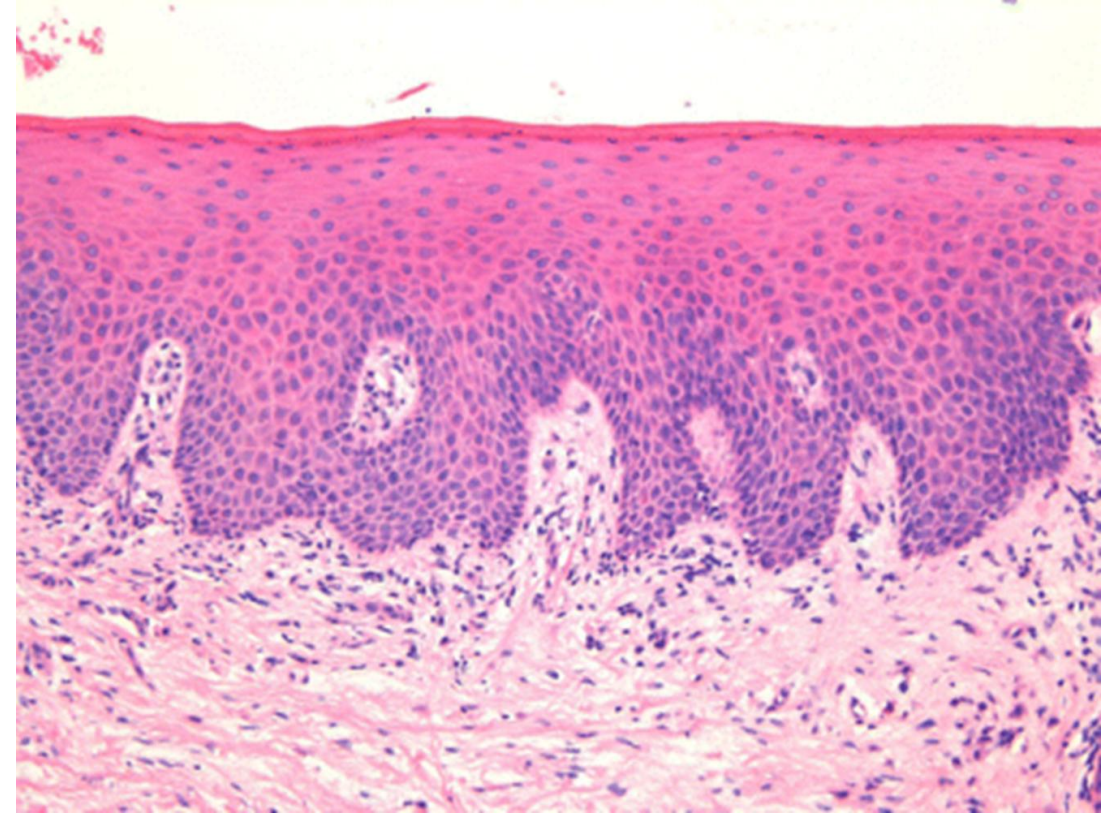
Lamina lucida
Lamina densa
Reticular lamina
Basal lamina

Periodic Acid-Schiff (PAS) stain : Basement membrane



BM contains **neutral mucopolysaccharides** (glycosaminoglycans). It stains positively with the **periodic acid-Schiff method**.

H &E stain : Basement membrane



CELLULAR ATTACHMENTS

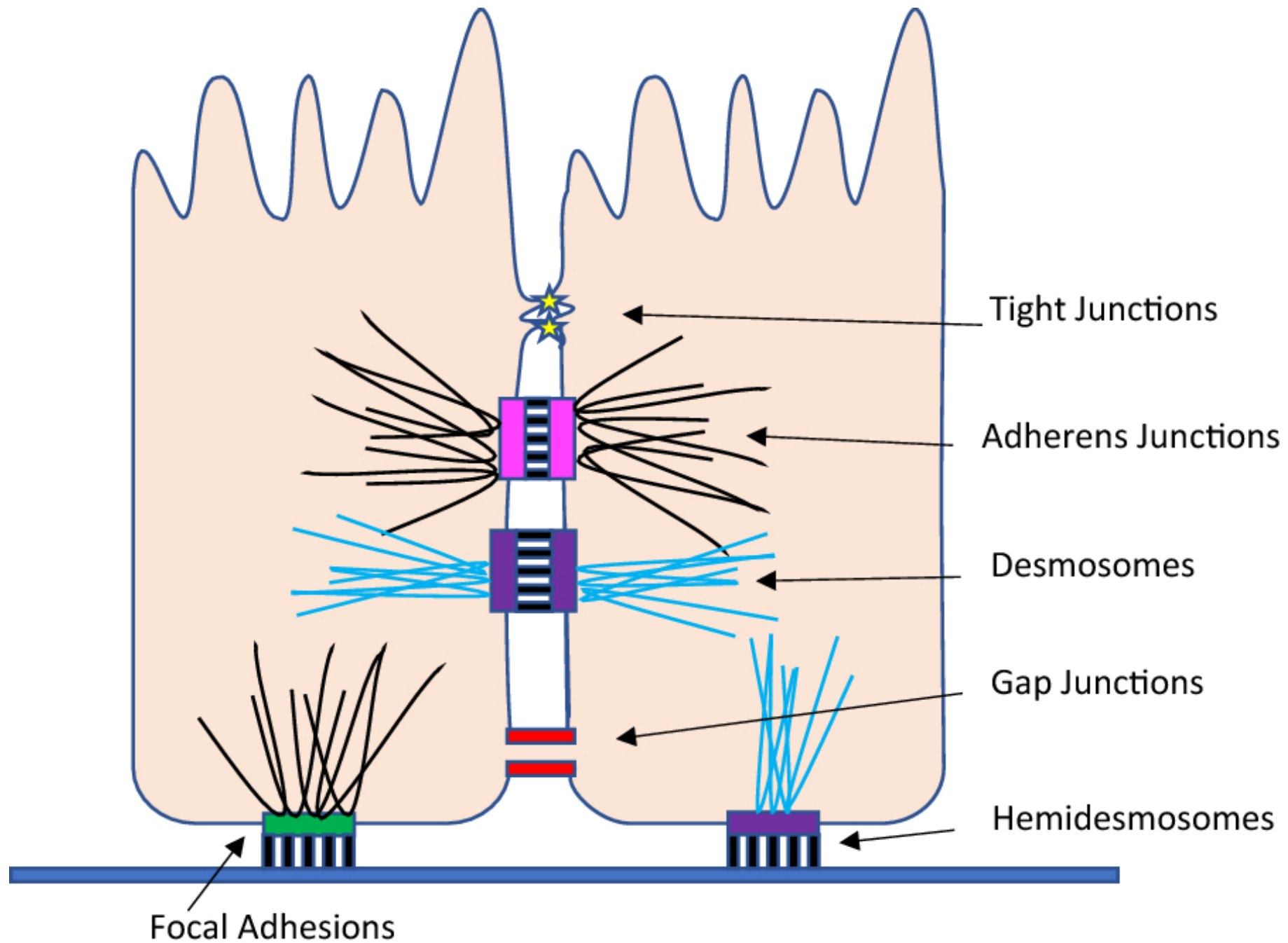
1. Desmosomes – Transmembrane glycoprotein: **CADHERINS(Desmoglein +desmocollin)**

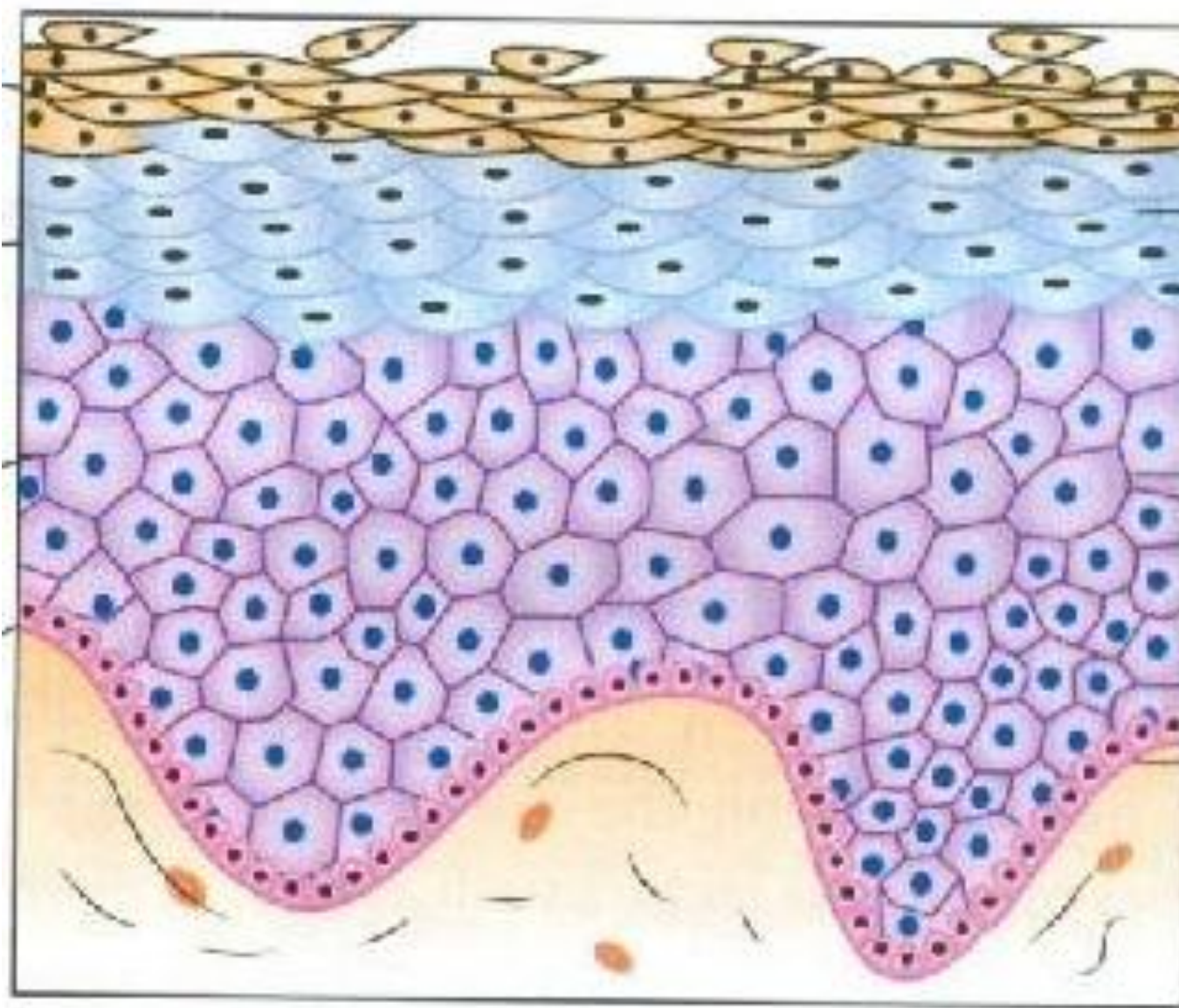
Attachment for epithelial cells in stratum spinosum, stratum basale and stratum granulosum

2. Hemidesmosomes – Transmembrane glycoprotein: **INTEGRINS**

attachment for basal cells of epithelium to basement membrane

Hemidesmosomes (HD) are **specialized junctional complexes**, that contribute to the attachment of **epithelial cells to the underlying basement membrane** in stratified and other complex epithelia, such as the skin.

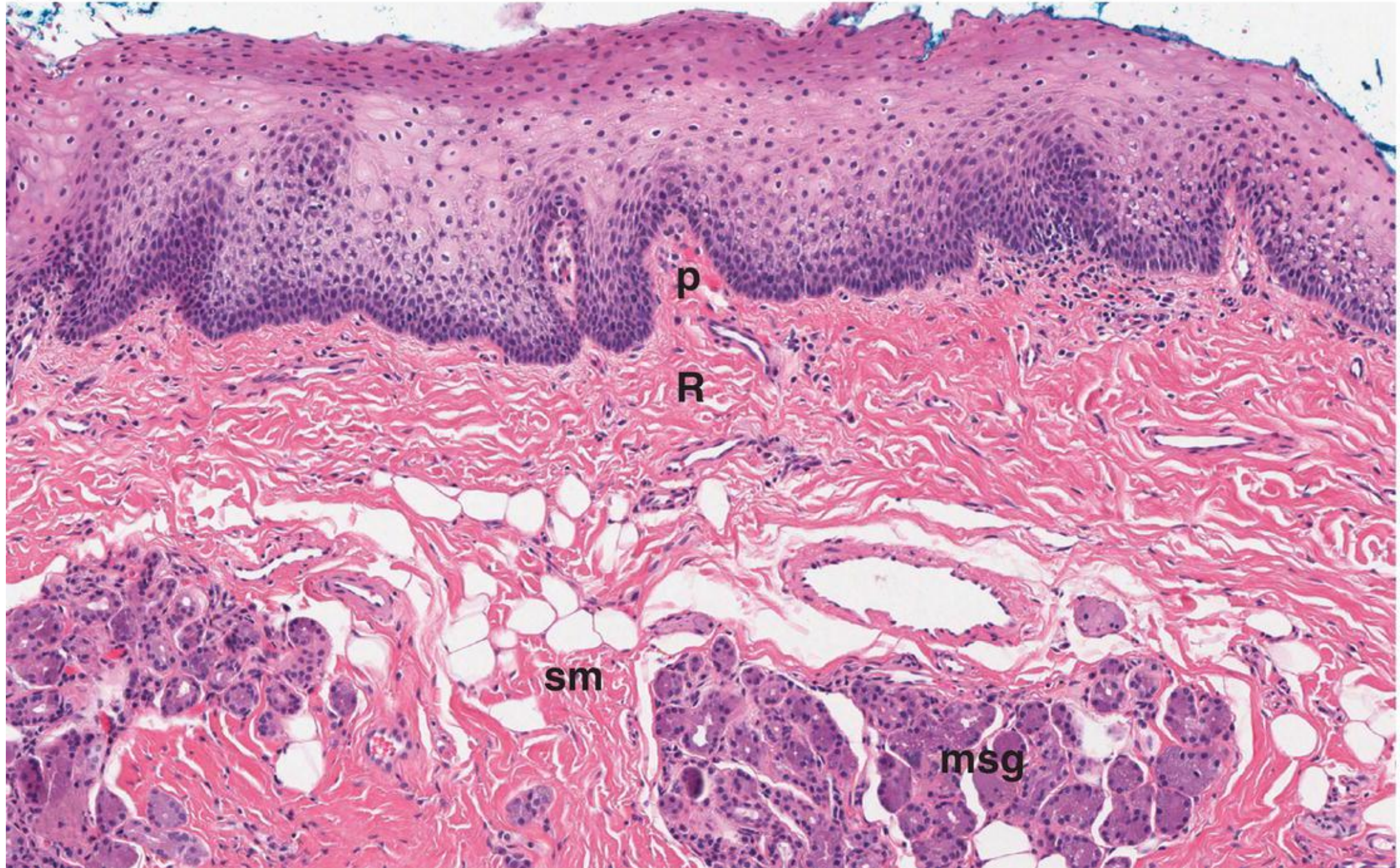




SUMMERY AND DISCUSSION

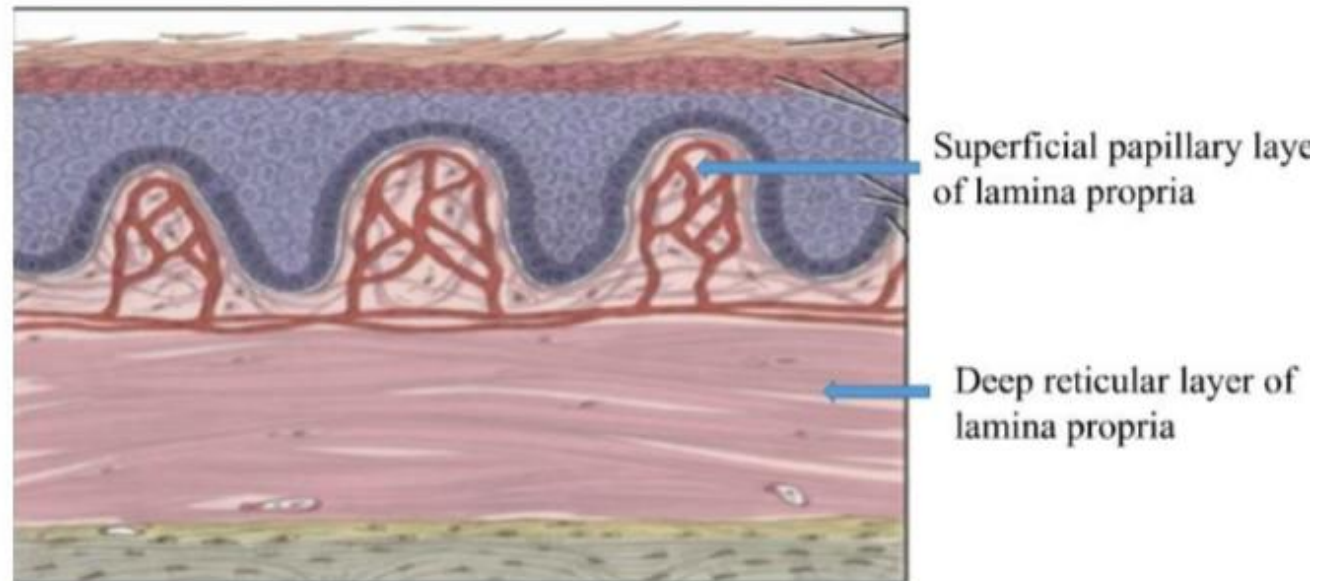
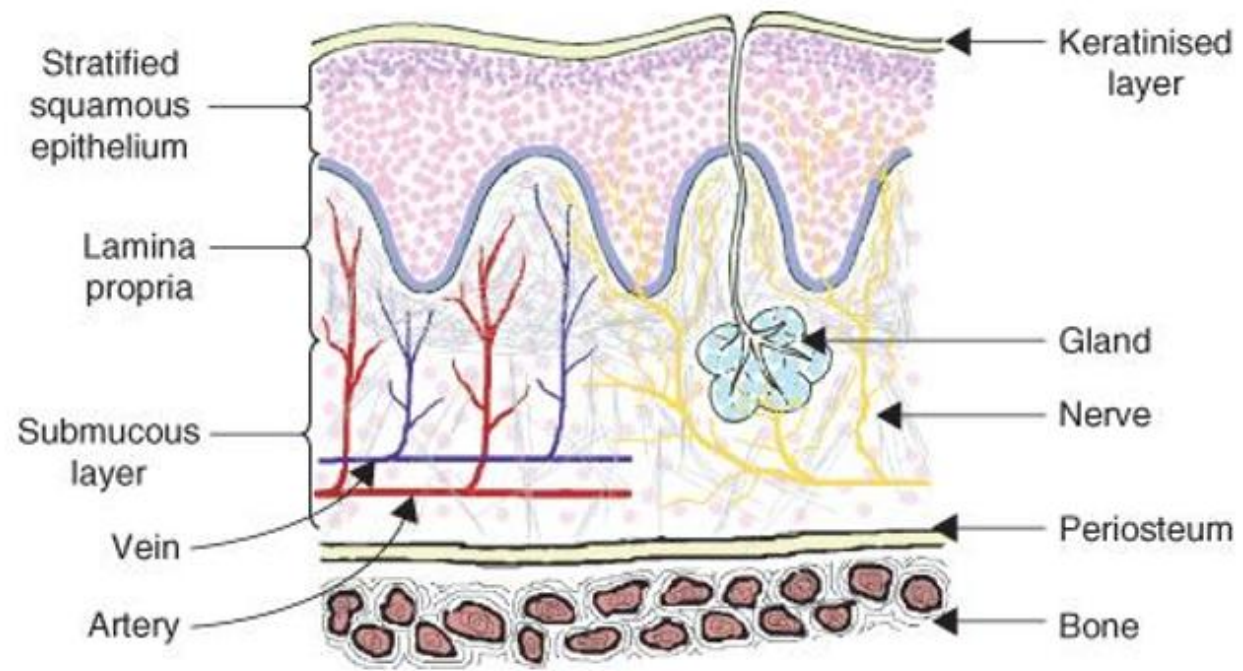


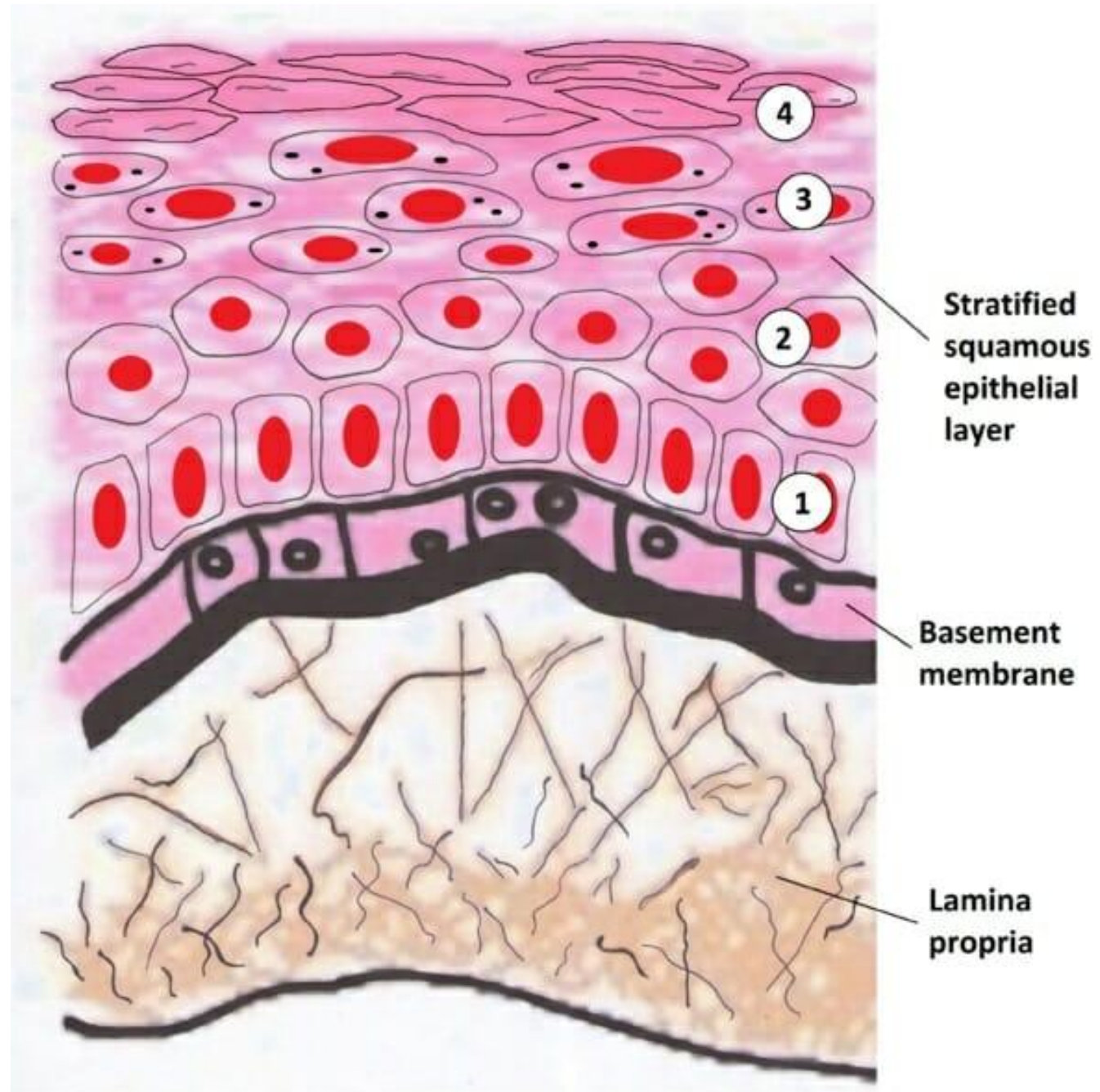
(a)



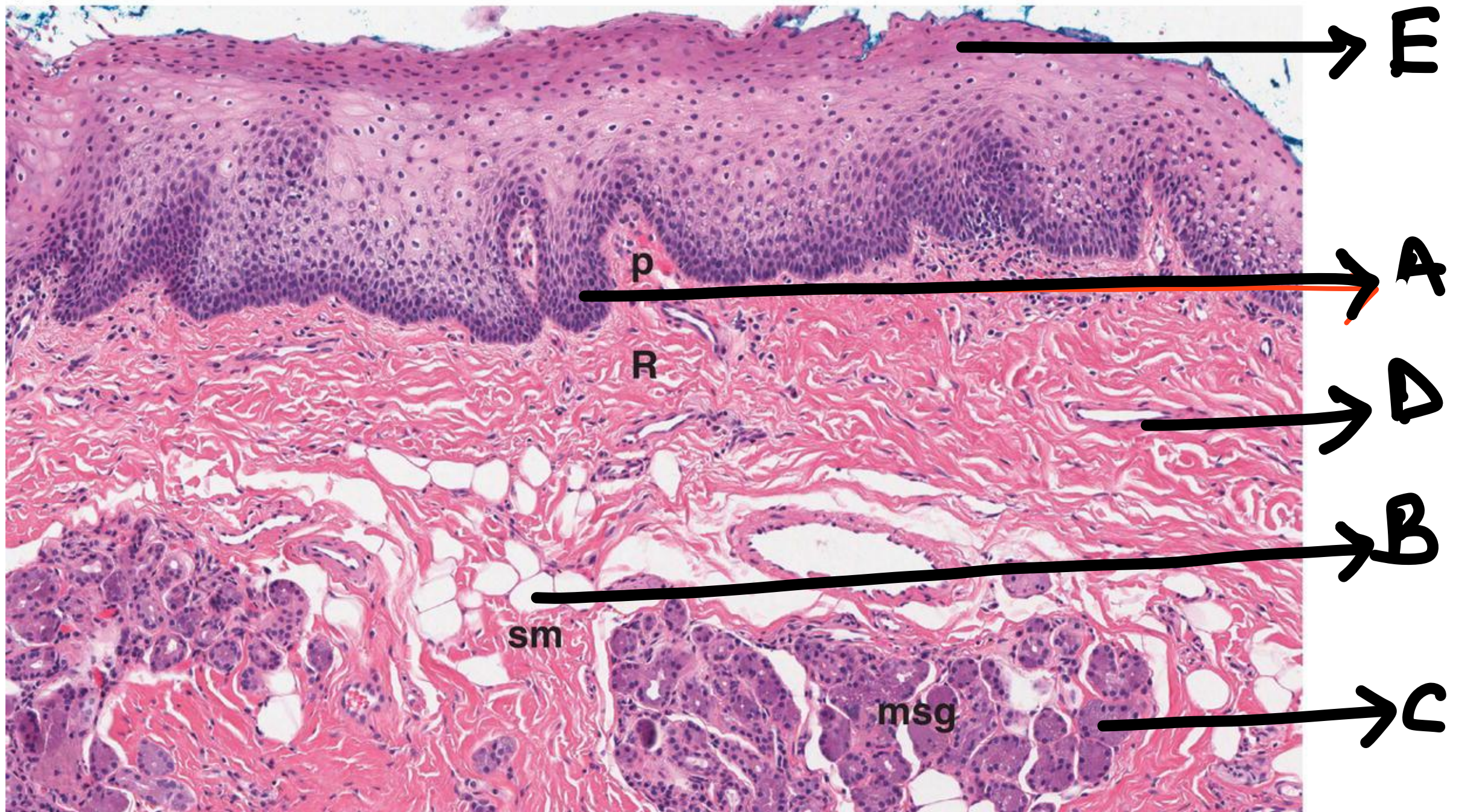
SUMMERY AND DISCUSSION



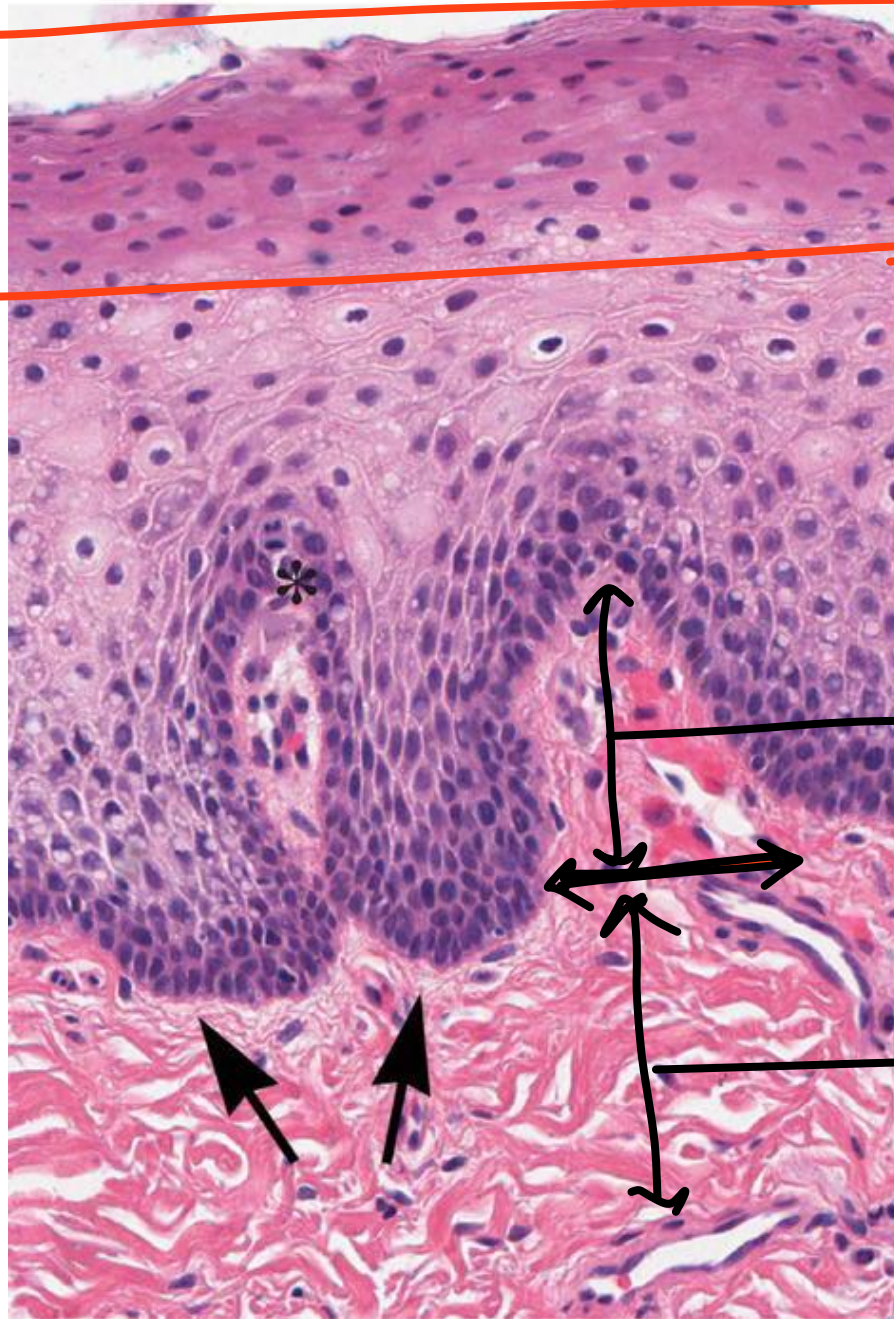




(a)



(b)



Type of
Keratinized
epithelium?

A Type of LP?

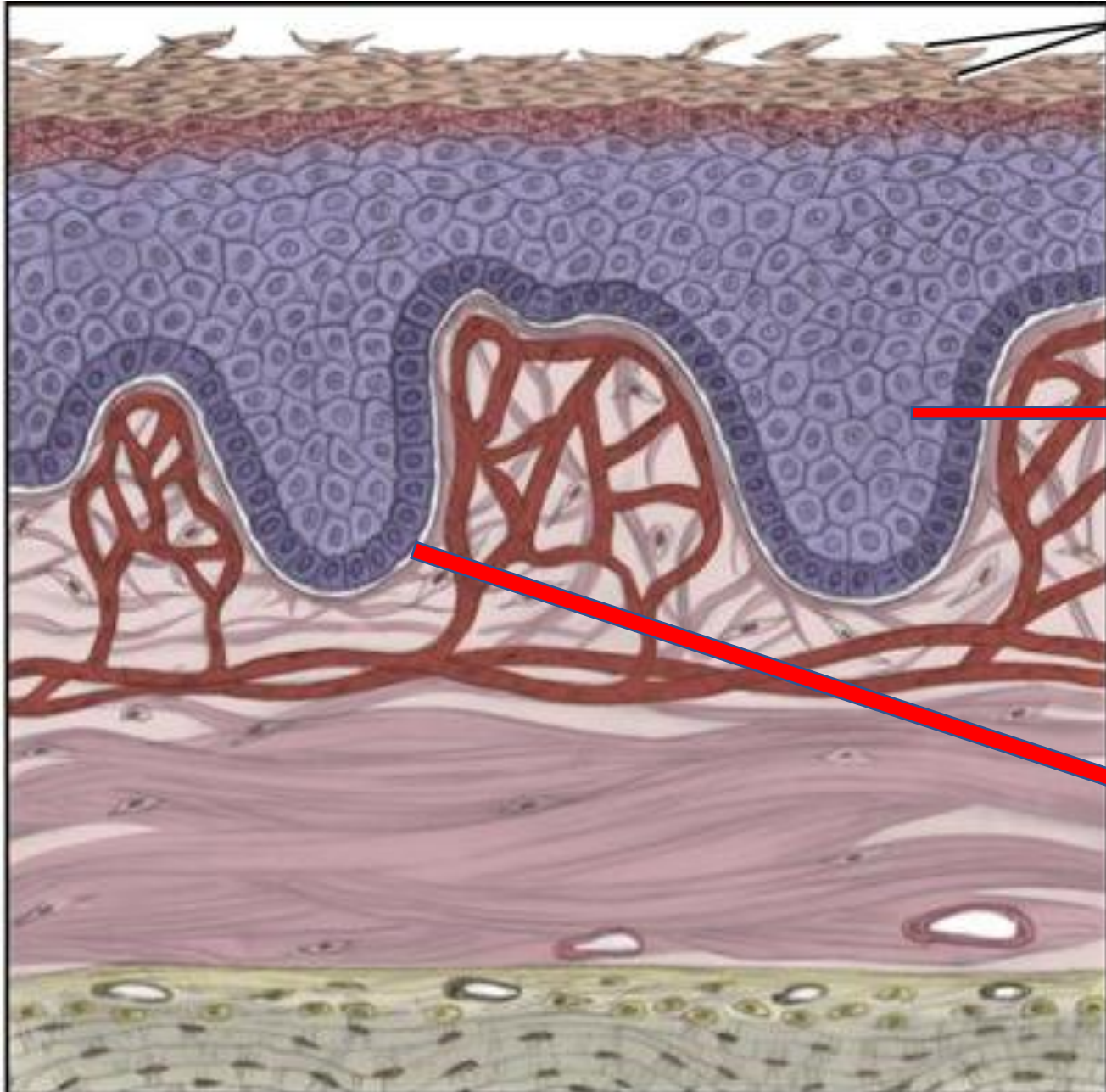
B Type of LP?

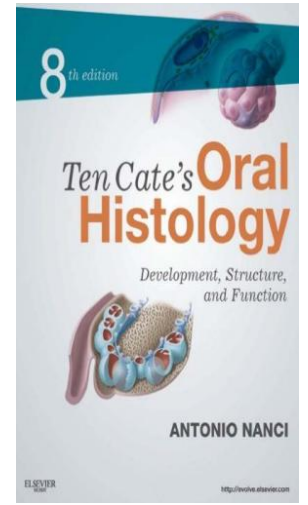
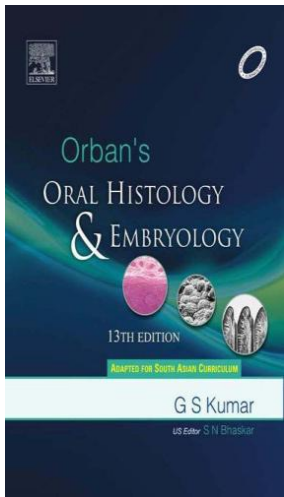
?

IDENTIFY TYPE OF JUNCTIONS

A

B





- Oral Development and Histology Avery, j. K.1st edition.
- Orban's Oral Histology and Embryology Bhaskar, s. N.11th edition.
- Oral Histology : Development, Structure and Funct Tencate, a. R. 4th edition.



Thank
you!!!

